13

Universal Gravitation

CHAPTER OUTLINE

13.1 Newton's Law of Universal Gravitation

13.2 Free-Fall Acceleration and the Gravitational Force

13.3 Analysis Model: Particle in a Field (Gravitational)

13.4 Kepler's Laws and the Motion of Planets

13.5 Gravitational Potential Energy

13.6 Energy Considerations in Planetary and Satellite Motion

* An asterisk indicates a question or problem new to this edition.

ANSWERS TO OBJECTIVE QUESTIONS

OQ13.1 Answer (c). Ten terms are needed in the potential energy:

$$U = U_{12} + U_{13} + U_{14} + U_{15} + U_{23} + U_{24} + U_{25} + U_{34} + U_{35} + U_{45}$$

OQ13.2 The ranking is a > b = c. The gravitational potential energy of the Earth-Sun system is negative and twice as large in magnitude as the kinetic energy of the Earth relative to the Sun. Then the total energy is negative and equal in absolute value to the kinetic energy.

OQ13.3 Answer (d). The satellite experiences a gravitational force, always directed toward the center of its orbit, and supplying the centripetal force required to hold it in its orbit. This force gives the satellite a centripetal acceleration, even if it is moving with constant angular speed. At each point on the circular orbit, the gravitational force is directed along a radius line of the path, and is perpendicular to the motion of the satellite, so this force does no work on the satellite.

OQ13.4 Answer (d). Having twice the mass would make the surface gravitational field two times larger. But the inverse square law says that having twice the radius would make the surface acceleration due to gravitation four times smaller. Altogether, g at the surface of B

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becomes $(2\ \text{m/s}^2)(2)/4 = 1\ \text{m/s}^2$.

OQ13.5 Answer (b). Switching off gravity would let the atmosphere evaporate away, but switching off the atmosphere has no effect on the planet's gravitational field.

OQ13.6 Answer (b). The mass of a spherical body of radius R and density ρ is $M = \rho V = \rho(4\pi R^3/3)$. The escape velocity from the surface of this body may then be written in either of the following equivalent forms:

$$v_{esc} = \sqrt{\frac{2GM}{R}} \quad \text{and } v_{esc} = \sqrt{\frac{2G}{R}\left(\frac{4\pi\rho R^3}{3}\right)} = \sqrt{\frac{8\pi\rho G R^2}{3}}$$

We see that the escape velocity depends on the three properties (mass, density, and radius) of the planet. Also, the weight of an object on the surface of the planet is $F_g = mg = GMm/R^2$, giving

$$g = GM/R^2 = \frac{G}{R^2}\left[\rho\left(\frac{4\pi R^3}{3}\right)\right] = \frac{4}{3}\pi\rho G R$$

The free-fall acceleration at the planet's surface then depends on the same properties as does the escape velocity. Changing the value of g would necessarily change the escape velocity. Of the listed quantities, the only one that does not affect the escape velocity is the mass of the object.

OQ13.7 (i) Answer (e). According to the inverse square law, $1/4^2 = 16$ times smaller.

(ii) Answer (c). $mv^2/r = GMm/r^2$ predicts that v is proportional to $(1/r)^{1/2}$, so it becomes $(1/4)^{1/2} = 1/2$ as large.

(iii) Answer (a). According to Kepler's third law, $(4^3)^{1/2} = 8$ times larger; also, the circumference is 4 times larger and the speed 1/2 as large: $4/(1/2) = 8$.

OQ13.8 Answer (b). The Earth is farthest from the sun around July 4 every year, when it is summer in the northern hemisphere and winter in the southern hemisphere. As described by Kepler's second law, this is when the planet is moving slowest in its orbit. Thus it takes more time for the planet to plod around the 180° span containing the minimum-speed point.

OQ13.9 The ranking is b > a > c = d > e. The force is proportional to the product of the masses and inversely proportional to the square of the separation distance, so we compute $m_1 m_2/r^2$ for each case: (a) $2\cdot 3/1^2 = 6$ (b) 18 (c) $18/4 = 4.5$ (d) 4.5 (e) $16/4 = 4$.

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OQ13.10 Answer (c). The International Space Station orbits just above the atmosphere, only a few hundred kilometers above the ground. This distance is small compared to the radius of the Earth, so the gravitational force on the astronaut is only slightly less than on the ground. We might *think* the gravitational force is zero or nearly zero, because the orbiting astronauts *appear* to be weightless. They and the space station are in free fall, so the normal force of the space station's wall/floor/ceiling on the astronauts is zero; they float freely around the cabin.

OQ13.11 Answer (e). We assume that the elliptical orbit is so elongated that the Sun, at one focus, is almost at one end of the major axis. If the period, T, is expressed in years and the semimajor axis, a, in astronomical units (AU), Kepler's third law states that $T^2 = a^3$. Thus, for Halley's comet, with a period of $T = 76$ y, the semimajor axis of its orbit is

$$a = \sqrt[3]{(76)^2} = 18 \text{ AU}$$

The length of the major axis, and the approximate maximum distance from the Sun, is $2a = 36$ AU.

ANSWERS TO CONCEPTUAL QUESTIONS

CQ13.1 (a) The gravitational force is conservative. (b) Yes. An encounter with a stationary mass cannot permanently speed up a spacecraft. But Jupiter is moving. A spacecraft flying across its orbit just behind the planet will gain kinetic energy because of the change in potential energy of the spacecraft-planet system. This is a collision because the spacecraft and planet exert forces on each other while they are isolated from outside forces. It is an elastic collision because only conservative forces are involved. (c) The planet loses kinetic energy as the spacecraft gains it.

CQ13.2 Cavendish determined G. Then from $g = \dfrac{GM}{R^2}$, one may determine the mass of the Earth. The term "weighed" is better expressed as "massed."

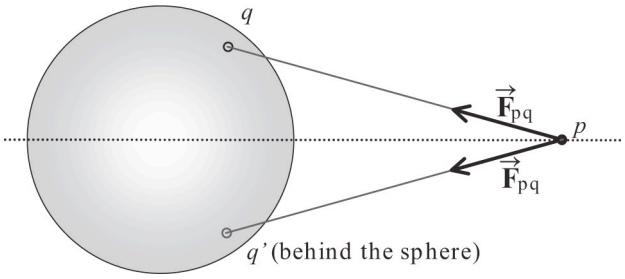
CQ13.3 For a satellite in orbit, one focus of an elliptical orbit, or the center of a circular orbit, must be located at the center of the Earth. If the satellite is over the northern hemisphere for half of its orbit, it must be over the southern hemisphere for the other half. We could share with Easter Island a satellite that would look straight down on Arizona each morning and vertically down on Easter Island each

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evening.

CQ13.4 (a) Every point q on the sphere that does not lie along the axis connecting the center of the sphere and the particle will have companion point q' for which the components of the gravitational force perpendicular to the axis will cancel. Point q' can be found by rotating the sphere through 180° about the axis.

(b) The forces will not necessarily cancel if the mass is not uniformly distributed, unless the center of mass of the nonuniform sphere still lies along the axis.



ANS. FIG. CQ13.4

CQ13.5 The angular momentum of a planet going around a sun is conserved. (a) The speed of the planet is maximum at closest approach. (b) The speed is a minimum at farthest distance. These two points, perihelion and aphelion respectively, are 180° apart, at opposite ends of the major axis of the orbit.

CQ13.6 Set the universal description of the gravitational force, $F_g = \dfrac{GM_X m}{R_X^2}$, equal to the local description, $F_g = ma_{\text{gravitational}}$, where M_x and R_x are the mass and radius of planet X, respectively, and m is the mass of a "test particle." Divide both sides by m.

CQ13.7 (a) In one sense, 'no'. If the object is at the very center of the Earth there is no other mass located there for comparison and the formula does not apply in the same way it was being applied while the object was some distance from the center. In another sense, 'yes'. One would have to compare, though, the distance between the object with mass m to the other individual masses that make up the Earth.

(b) The gravitational force of the Earth on an object at its center must be zero, not infinite as one interpretation of Equation 11.1 would suggest. All the bits of matter that make up the Earth pull in different outward directions on the object, causing the net force on it to be zero.

CQ13.8 The escape speed from the Earth is 11.2 km/s and that from the

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Moon is 2.3 km/s, smaller by a factor of 5. The energy required—and fuel—would be proportional to v^2, or 25 times more fuel is required to leave the Earth versus leaving the Moon.

CQ13.9 Air resistance causes a decrease in the energy of the satellite-Earth system. This reduces the radius of the orbit, bringing the satellite closer to the surface of the Earth. A satellite in a smaller orbit, however, must travel faster. Thus, the effect of air resistance is to speed up the satellite!

SOLUTIONS TO END-OF-CHAPTER PROBLEMS

Section 13.1 Newton's Law of Universal Gravitation

P13.1 This is a direct application of the equation expressing Newton's law of gravitation:

$$F = \frac{GMm}{r^2} = \left(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2\right)\frac{\left(1.50\ \text{kg}\right)\left(15.0 \times 10^{-3}\ \text{kg}\right)}{\left(4.50 \times 10^{-2}\ \text{m}\right)^2}$$

$$= \boxed{7.41 \times 10^{-10}\ \text{N}}$$

P13.2 For two 70-kg persons, modeled as spheres,

$$F_g = \frac{Gm_1m_2}{r^2} = \frac{\left(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2\right)\left(70\ \text{kg}\right)\left(70\ \text{kg}\right)}{\left(2\ \text{m}\right)^2}$$

$$\boxed{\sim 10^{-7}\ \text{N}}$$

P13.3 (a) At the midpoint between the two objects, the forces exerted by the 200-kg and 500-kg objects are oppositely directed,

and from $F_g = \dfrac{Gm_1m_2}{r^2}$

we have $$\sum F = \frac{G\left(50.0\ \text{kg}\right)\left(500\ \text{kg} - 200\ \text{kg}\right)}{\left(2.00\ \text{m}\right)^2} = \boxed{2.50 \times 10^{-7}\ \text{N}}$$

toward the 500-kg object.

(b) At a point between the two objects at a distance d from the 500-kg object, the net force on the 50.0-kg object will be zero when

$$\frac{G\left(50.0\ \text{kg}\right)\left(200\ \text{kg}\right)}{\left(4.00\ \text{m} - d\right)^2} = \frac{G\left(50.0\ \text{kg}\right)\left(500\ \text{kg}\right)}{d^2}$$

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To solve, cross-multiply to clear of fractions and take the square root of both sides. The result is

$d = \boxed{2.45 \text{ m from the 500-kg object toward the smaller object}}$.

P13.4 (a) The Sun-Earth distance is 1.496×10^{11} m and the Earth-Moon distance is 3.84×10^{8} m, so the distance from the Sun to the Moon during a solar eclipse is

$$1.496 \times 10^{11} \text{ m} - 3.84 \times 10^{8} \text{ m} = 1.492 \times 10^{11} \text{ m}$$

The mass of the Sun, Earth, and Moon are

$$M_S = 1.99 \times 10^{30} \text{ kg}$$

$$M_E = 5.98 \times 10^{24} \text{ kg}$$

and $M_M = 7.36 \times 10^{22}$ kg

We have

$$F_{SM} = \frac{Gm_1m_2}{r^2}$$

$$= \frac{\left(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2\right)\left(1.99 \times 10^{30} \text{ kg}\right)\left(7.36 \times 10^{22} \text{ kg}\right)}{\left(1.492 \times 10^{11} \text{ m}\right)^2}$$

$$= \boxed{4.39 \times 10^{20} \text{ N}}$$

(b) $$F_{EM} = \frac{\left(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2\right)\left(5.98 \times 10^{24} \text{ kg}\right)\left(7.36 \times 10^{22} \text{ kg}\right)}{\left(3.84 \times 10^{8} \text{ m}\right)^2}$$

$$= \boxed{1.99 \times 10^{20} \text{ N}}$$

(c) $$F_{SE} = \frac{\left(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2\right)\left(1.99 \times 10^{30} \text{ kg}\right)\left(5.98 \times 10^{24} \text{ kg}\right)}{\left(1.496 \times 10^{11} \text{ m}\right)^2}$$

$$= \boxed{3.55 \times 10^{22} \text{ N}}$$

(d) The force exerted by the Sun on the Moon is much stronger than the force of the Earth on the Moon. In a sense, the Moon orbits the Sun more than it orbits the Earth. The Moon's path is everywhere concave toward the Sun. Only by subtracting out the solar orbital motion of the Earth-Moon system do we see the Moon orbiting the center of mass of this system.

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P13.5 With one metric ton = 1 000 kg,

$$F = m_1 g = \frac{G m_1 m_2}{r^2}$$

$$g = \frac{G m_2}{r^2} = \frac{\left(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2\right)\left(4.00 \times 10^{7}\ \text{kg}\right)}{(100\ \text{m})^2}$$

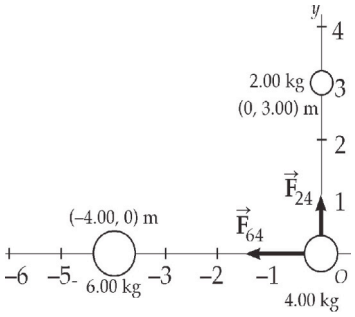
$$= \boxed{2.67 \times 10^{-7}\ \text{m/s}^2}$$

P13.6 The force exerted on the 4.00-kg mass by the 2.00-kg mass is directed upward and given by

$$\vec{\mathbf{F}}_{12} = G\frac{m_2 m_1}{r_{12}^2}\hat{\mathbf{j}}$$

$$= \left(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2\right) \frac{(4.00\ \text{kg})(2.00\ \text{kg})}{(3.00\ \text{m})^2}\hat{\mathbf{j}}$$

$$= 5.93 \times 10^{-11}\hat{\mathbf{j}}\ \text{N}$$



ANS. FIG. P13.6

The force exerted on the 4.00-kg mass by the 6.00-kg mass is directed to the left:

$$\vec{\mathbf{F}}_{32} = G\frac{m_2 m_3}{r_{32}^2}\left(-\hat{\mathbf{i}}\right)$$

$$= \left(-6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2\right)\frac{(4.00\ \text{kg})(6.00\ \text{kg})}{(4.00\ \text{m})^2}\hat{\mathbf{i}}$$

$$= -10.0 \times 10^{-11}\hat{\mathbf{i}}\ \text{N}$$

Therefore, the resultant force on the 4.00-kg mass is

$$\vec{\mathbf{F}}_4 = \vec{\mathbf{F}}_{24} + \vec{\mathbf{F}}_{64} = \boxed{\left(-10.0\hat{\mathbf{i}} + 5.93\hat{\mathbf{j}}\right) \times 10^{-11}\ \text{N}}$$

***P13.7** The magnitude of the gravitational force is given by

$$F = \frac{G m_1 m_2}{r^2} = \frac{\left(6.672 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2\right)(2.00\ \text{kg})(2.00\ \text{kg})}{(0.300\ \text{m})^2}$$

$$= \boxed{2.97 \times 10^{-9}\ \text{N}}$$

P13.8 Assume the masses of the sphere are the same. Using $F_g = \frac{G m_1 m_2}{r^2}$, we would find that the mass of a sphere is 1.22×10^5 kg! If the spheres have at most a radius of 0.500 m, the density of spheres would be at

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least 2.34×10^5 kg/m^3, which is ten times the density of the most dense element, osmium.

The situation is impossible because no known element could compose the spheres.

P13.9 We are given $m_1 + m_2 = 5.00$ kg, which means that $m_2 = 5.00 \text{ kg} - m_1$. Newton's law of universal gravitation then becomes

$$F = G\frac{m_1 m_2}{r^2}$$

$$\Rightarrow 1.00 \times 10^{-8} \text{ N} = \left(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2\right)\frac{m_1\left(5.00 \text{ kg} - m_1\right)}{(0.200 \text{ m})^2}$$

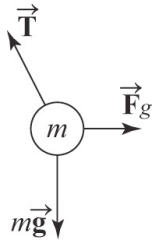
$$\left(5.00 \text{ kg}\right)m_1 - m_1^2 = \frac{\left(1.00 \times 10^{-8} \text{ N}\right)\left(0.040\,0 \text{ m}^2\right)}{6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2} = 6.00 \text{ kg}^2$$

Thus, $m_1^2 - \left(5.00 \text{ kg}\right)m_1 + 6.00 \text{ kg} = 0$

or $\left(m_1 - 3.00 \text{ kg}\right)\left(m_1 - 2.00 \text{ kg}\right) = 0$

giving $\boxed{m_1 = 3.00 \text{ kg, so } m_2 = 2.00 \text{ kg}}$. The answer $m_1 = 2.00$ kg and $m_2 = 3.00$ kg is physically equivalent.

P13.10 Let θ represent the angle each cable makes with the vertical, L the cable length, x the distance each ball is displaced by the gravitational force, and $d = 1$ m the original distance between them. Then $r = d - 2x$ is the separation of the balls. We have



ANS. FIG. P13.10

$$\sum F_y = 0: \quad T\cos\theta - mg = 0$$

$$\sum F_x = 0: \quad T\sin\theta - \frac{Gmm}{r^2} = 0$$

Then $\tan\theta = \dfrac{Gmm}{r^2 mg}$

$$\frac{x}{\sqrt{L^2 - x^2}} = \frac{Gm}{g(d-2x)^2} \quad \rightarrow \quad x(d-2x)^2 = \frac{Gm}{g}\sqrt{L^2 - x^2}$$

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The factor $\frac{Gm}{g}$ is numerically small. We expect that x is very small compared to both L and d, so we can treat the term $(d - 2x)$ as d, and $(L^2 - x^2)$ as L^2. We then have

$$x(1\text{ m})^2 = \frac{\left(6.67\times10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2\right)(100\text{ kg})}{\left(9.80\text{ m/s}^2\right)}(45.00\text{ m})$$

$$\boxed{x = 3.06\times10^{-8}\text{ m}}$$

Section 13.2 Free-Fall Acceleration and the Gravitational Force

P13.11 The distance of the meteor from the center of Earth is $R + 3R = 4R$. Calculate the acceleration of gravity at this distance.

$$g = \frac{GM}{r^2} = \frac{(6.67\times10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\text{ kg})}{[4(6.37\times10^{6}\text{ m})]^2}$$

$$= \boxed{0.614\text{ m/s}^2\text{, toward Earth}}$$

P13.12 The gravitational field at the surface of the Earth or Moon is given by $g = \frac{GM}{R^2}$.

The expression for density is $\rho = \frac{M}{V} = \frac{M}{\frac{4}{3}\pi R^3}$,

so $$M = \frac{4}{3}\pi\rho R^3$$

and $$g = \frac{G\left(\frac{4}{3}\pi\rho R^3\right)}{R^2} = \frac{4}{3}G\pi\rho R$$

Noting that this equation applies to both the Moon and the Earth, and dividing the two equations,

$$\frac{g_M}{g_E} = \frac{\frac{4}{3}G\pi\rho_M R_M}{\frac{4}{3}G\pi\rho_E R_E} = \frac{\rho_M R_M}{\rho_E R_E}$$

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Substituting for the fractions,

$$\frac{1}{6} = \frac{\rho_M}{\rho_E}\left(\frac{1}{4}\right) \quad \text{and} \quad \frac{\rho_M}{\rho_E} = \frac{4}{6} = \boxed{\frac{2}{3}}$$

P13.13 (a) For the gravitational force on an object in the neighborhood of Miranda, we have

$$m_{\text{obj}} g = \frac{G m_{\text{obj}} m_{\text{Miranda}}}{r_{\text{Miranda}}^2}$$

$$g = \frac{G m_{\text{Miranda}}}{r_{\text{Miranda}}^2} = \frac{\left(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2 / \text{kg}^2\right)\left(6.68 \times 10^{19}\ \text{kg}\right)}{\left(242 \times 10^3\ \text{m}\right)^2}$$

$$= \boxed{0.076\,1\ \text{m/s}^2}$$

(b) We ignore the difference (of about 4%) in g between the lip and the base of the cliff. For the vertical motion of the athlete, we have

$$y_f = y_i + v_{yi} + \frac{1}{2} a_y t^2$$

$$-5\ 000\ \text{m} = 0 + 0 + \frac{1}{2}\left(-0.076\ 1\ \text{m/s}^2\right) t^2$$

$$t = \left(\frac{2\left(5\ 000\ \text{m}\right)\text{s}^2}{0.076\ 1\ \text{m}}\right)^{1/2} = \boxed{363\ \text{s}}$$

(c) $$x_f = x_i + v_{xi} t + \frac{1}{2} a_x t^2 = 0 + (8.50\ \text{m/s})(363\ \text{s}) + 0 = \boxed{3.08 \times 10^3\ \text{m}}$$

We ignore the curvature of the surface (of about 0.7°) over the athlete's trajectory.

(d) $$v_{xf} = v_{xi} = 8.50\ \text{m/s}$$

$$v_{yf} = v_{yi} + a_y t = 0 - \left(0.076\,1\ \text{m/s}^2\right)(363\ \text{s}) = -27.6\ \text{m/s}$$

Thus $\vec{\mathbf{v}}_f = \left(8.50\hat{\mathbf{i}} - 27.6\hat{\mathbf{j}}\right)\ \text{m/s} = \sqrt{8.50^2 + 27.6^2}\ \text{m/s}$ at

$\tan^{-1}\left(\dfrac{27.6\ \text{m/s}}{8.50\ \text{m/s}}\right) = 72.9°$ below the x axis.

$$\boxed{\vec{\mathbf{v}}_f = 28.9\ \text{m/s at } 72.9° \text{ below the horizontal}}$$

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Section 13.3 Analysis Model: Particle in a Field (Gravitational)

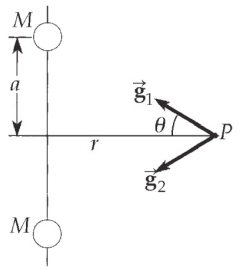
P13.14 (a) $g_1 = g_2 = \dfrac{MG}{r^2 + a^2}$

$g_{1y} = -g_{2y}$

$g_y = g_{1y} + g_{2y} = 0$

$g_{1x} = g_{2x} = g_2 \cos\theta$

$\cos\theta = \dfrac{r}{\left(a^2 + r^2\right)^{1/2}}$



ANS. FIG. P13.14

$\vec{\mathbf{g}} = 2g_{2x}\left(-\hat{\mathbf{i}}\right)$

or $\vec{\mathbf{g}} = \boxed{\dfrac{2MGr}{\left(r^2 + a^2\right)^{3/2}} \text{ toward the center of mass}}$

(b) At $r = 0$, the fields of the two objects are equal in magnitude and opposite in direction, to add to zero.

(c) As $r \to 0$, $2MGr(r^2 + a^2)^{-3/2}$ approaches $2MG(0)/a^3 = 0$.

(d) When r is much greater than a, the angles the field vectors make with the x axis become smaller. At very great distances, the field vectors are almost parallel to the axis; therefore, they begin to look like the field vector from a single object of mass $2M$.

(e) As r becomes much larger than a, the expression approaches $2MGr(r^2 + 0^2)^{-3/2} = 2MGr/r^3 = 2MG/r^2$ as required.

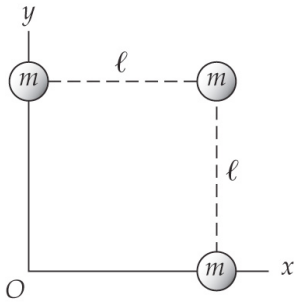
P13.15 The vector gravitational field at point O is given by

$$\vec{\mathbf{g}} = \frac{Gm}{l^2}\hat{\mathbf{i}} + \frac{Gm}{l^2}\hat{\mathbf{j}} + \frac{Gm}{2l^2}\left(\cos 45.0°\,\hat{\mathbf{i}} + \sin 45.0\hat{\mathbf{j}}\right)$$

so $$\vec{\mathbf{g}} = \frac{Gm}{l^2}\left(1 + \frac{1}{2\sqrt{2}}\right)\left(\hat{\mathbf{i}} + \hat{\mathbf{j}}\right) \text{ or}$$

$$\boxed{\vec{\mathbf{g}} = \frac{Gm}{l^2}\left(\sqrt{2} + \frac{1}{2}\right) \text{toward the opposite corner.}}$$

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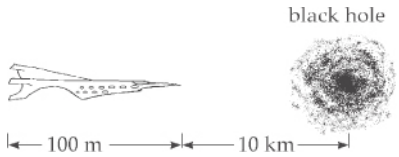


ANS. FIG. P13.15

P13.16 (a) $$F = \frac{GMm}{r^2} = \frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left[100\left(1.99\times10^{30}\ \text{kg}\right)\left(10^3\ \text{kg}\right)\right]}{\left(1.00\times10^4\ \text{m}+50.0\ \text{m}\right)^2}$$

$$= \boxed{1.31\times10^{17}\ \text{N}}$$

(b) $$\Delta F = \frac{GMm}{r_{\text{front}}^2} - \frac{GMm}{r_{\text{back}}^2}$$



ANS. FIG. P13.16

$$\Delta g = \frac{\Delta F}{m} = \frac{GM\left(r_{\text{back}}^2 - r_{\text{front}}^2\right)}{r_{\text{front}}^2 r_{\text{back}}^2}$$

$$\Delta g = \left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)$$
$$\frac{\left[100\left(1.99\times10^{30}\ \text{kg}\right)\right]\left[\left(1.01\times10^4\ \text{m}\right)^2 - \left(1.00\times10^4\ \text{m}\right)^2\right]}{\left(1.00\times10^4\ \text{m}\right)^2\left(1.01\times10^4\ \text{m}\right)^2}$$

$$\Delta g = \boxed{2.62\times10^{12}\ \text{N/kg}}$$

Section 13.4 Kepler's Laws and the Motion of Planets

P13.17 The gravitational force on mass located at distance r from the center of the Earth is $F_g = mg = GM_E m/r^2$. Thus, the acceleration of gravity at this location is $g = GM_E/r^2$. If $g = 9.00\ \text{m/s}^2$ at the location of the satellite, the radius of its orbit must be

$$r = \sqrt{\frac{GM_E}{g}} = \sqrt{\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)}{9.00\ \text{m/s}^2}}$$

$$= 6.66\times10^6\ \text{m}$$

From Kepler's third law for Earth satellites, $T^2 = 4\pi^2 r^3 GM_E S$, the period is found to be

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$$T = 2\pi\sqrt{\frac{r^3}{GM_E}} = 2\pi\sqrt{\frac{\left(6.66\times10^{6}\ \text{m}\right)^3}{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)}}$$

$$= 5.41\times10^{3}\ \text{s}$$

or

$$T = \left(5.41\times10^{3}\ \text{s}\right)\left(\frac{1\ \text{h}}{3\,600\ \text{s}}\right) = \boxed{1.50\ \text{h} = 90.0\ \text{min}}$$

P13.18 The gravitational force exerted by Jupiter on Io causes the centripetal acceleration of Io. A force diagram of the satellite would show one downward arrow.

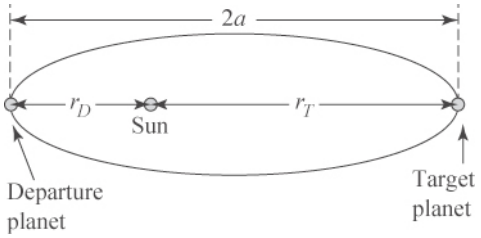
$$\sum F_{\text{on Io}} = M_{\text{Io}}a: \quad \frac{GM_{\text{J}}M_{\text{Io}}}{r^2} = \frac{M_{\text{Io}}v^2}{r} = \frac{M_{\text{Io}}}{r}\left(\frac{2\pi r}{T}\right)^2 = \frac{4\pi^2 r M_{\text{Io}}}{T^2}$$

Thus the mass of Io divides out and we have Kepler's third law with $m << M$,

$$M_J = \frac{4\pi^2 r^3}{GT^2} = \frac{4\pi^2(4.22\times10^{8}\,\text{m})^3}{(6.67\times10^{-11}\text{N}\cdot\text{m}^2/\text{kg}^2)(1.77\ \text{d})^2}\left(\frac{1\ \text{d}}{86\,400\ \text{s}}\right)^2$$

and $M_J = \boxed{1.90\times10^{27}\ \text{kg}}$ (approximately 316 Earth masses)

P13.19 (a) The desired path is an elliptical trajectory with the Sun at one of the foci, the departure planet at the perihelion, and the target planet at the aphelion. The perihelion distance r_D is the radius of the departure planet's orbit, while the aphelion distance r_T is the radius of the target planet's orbit. The semimajor axis of the desired trajectory is then $a = \left(r_D + r_T\right)/2$.



ANS. FIG. P13.19

If Earth is the departure planet, $r_D = 1.496\times10^{11}\ \text{m} = 1.00\ \text{AU}$

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With Mars as the target planet,

$$r_T = 2.28\times10^{11}\text{ m}\left(\frac{1\text{ AU}}{1.496\times10^{11}\text{ m}}\right) = 1.52\text{ AU}$$

Thus, the semimajor axis of the minimum energy trajectory is

$$a = \frac{r_D + r_T}{2} = \frac{1.00\text{ AU} + 1.52\text{ AU}}{2} = 1.26\text{ AU}$$

Kepler's third law, $T^2 = a^3$, then gives the time for a full trip around this path as

$$T = \sqrt{a^3} = \sqrt{(1.26\text{ AU})^3} = 1.41\text{ yr}$$

so the time for a one-way trip from Earth to Mars is

$$\Delta t = \frac{1}{2}T = \frac{1.41\text{ yr}}{2} = \boxed{0.71\text{ yr}}$$

(b) This trip cannot be taken at just any time. The departure must be timed so that the spacecraft arrives at the aphelion when the target planet is located there.

P13.20 (a) The particle does possess angular momentum, because it is not headed straight for the origin.

(b) Its angular momentum is constant. There are no identified outside influences acting on the object.

(c) Since speed is constant, the distance traveled between t_A and t_B is equal to the distance traveled between t_C and t_D. The area of a triangle is equal to one-half its (base) width across one side times its (height) dimension perpendicular to that side.

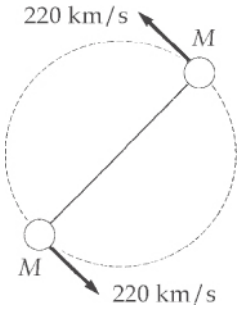
So $\frac{1}{2}bv_0(t_B - t_A) = \frac{1}{2}bv_0(t_D - t_C)$

states that the particle's radius vector sweeps out equal areas in equal times.

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P13.21 Applying Newton's second law, $\sum F = ma$ yields $F_g = ma_c$ for each star:

$$\frac{GMM}{(2r)^2} = \frac{Mv^2}{r} \qquad \text{or} \qquad M = \frac{4v^2 r}{G}$$



ANS. FIG. P13.21

We can write r in terms of the period, T, by considering the time and distance of one complete cycle. The distance traveled in one orbit is the circumference of the stars' common orbit, so $2\pi r = vT$. Therefore,

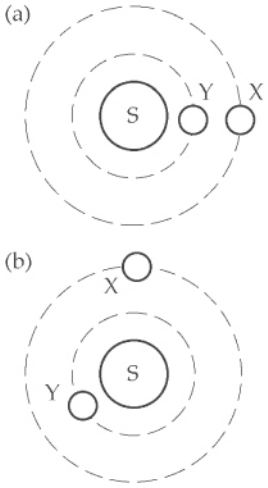
$$M = \frac{4v^2 r}{G} = \frac{4v^2}{G}\left(\frac{vT}{2\pi}\right)$$

so,
$$M = \frac{2v^3 T}{\pi G} = \frac{2\left(220\times10^3\ \text{m/s}\right)^3 (14.4\ \text{d})(86\,400\ \text{s/d})}{\pi\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)}$$

$$= \boxed{1.26\times10^{32}\ \text{kg} = 63.3\ \text{solar masses}}$$

P13.22 To find the angular displacement of planet Y, we apply Newton's second law:

$$\sum F = ma: \quad \frac{Gm_{\text{planet}}M_{\text{star}}}{r^2} = \frac{m_{\text{planet}}v^2}{r}$$



ANS. FIG. P13.22

Then, using $v = r\omega$,

$$\frac{GM_{\text{star}}}{r} = v^2 = r^2\omega^2$$

$$GM_{\text{star}} = r^3\omega^2 = r_x^3\omega_x^2 = r_y^3\omega_y^2$$

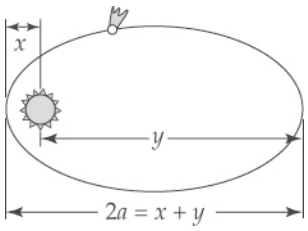
solving for the angular velocity of planet Y gives

$$\omega_y = \omega_x\left(\frac{r_x}{r_y}\right)^{3/2} = \left(\frac{90.0°}{5.00\,\text{yr}}\right)3^{3/2} = \frac{468°}{5.00\,\text{yr}}$$

So, given that there are 360° in one revolution we convert 468° to find that planet *Y* has turned through $\boxed{1.30\ \text{revolutions}}$.

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P13.23 By Kepler's third law, $T^2 = ka^3$ (a = semimajor axis). For any object orbiting the Sun, with T in years and a in AU, $k = 1.00$. Therefore, for Comet Halley, and suppressing units,



ANS. FIG. P13.23

$$(75.6)^2 = (1.00)\left(\frac{0.570 + y}{2}\right)^3$$

The farthest distance the comet gets from the Sun is

$$y = 2(75.6)^{2/3} - 0.570 = \boxed{35.2 \text{ AU}}$$

(out around the orbit of Pluto).

***P13.24** By conservation of angular momentum for the satellite, $r_p v_p = r_a v_a$, or

$$\frac{v_p}{v_a} = \frac{r_a}{r_p} = \frac{2\,289 \text{ km} + 6.37 \times 10^3 \text{ km}}{459 \text{ km} + 6.37 \times 10^3 \text{ km}} = \frac{8\,659 \text{ km}}{6\,829 \text{ km}} = \boxed{1.27}$$

We do not need to know the period.

P13.25 For an object in orbit about Earth, Kepler's third law gives the relation between the orbital period T and the average radius of the orbit ("semimajor axis") as

$$T^2 = \left(\frac{4\pi^2}{GM_E}\right) r^3$$

We assume that the two given distances in the problem statements are the perigee and apogee, respectively.

Thus, if the average radius is

$$r = \frac{r_{\min} + r_{\max}}{2} = \frac{6\,670 \text{ km} + 385\,000 \text{ km}}{2}$$
$$= 1.96 \times 10^5 \text{ km} = 1.96 \times 10^8 \text{ m}$$

The period (time for a round trip from Earth to the Moon) would be

$$T = 2\pi\sqrt{\frac{r^3}{GM_E}}$$
$$= 2\pi\sqrt{\frac{\left(1.96 \times 10^8 \text{ m}\right)^3}{\left(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2\right)\left(5.98 \times 10^{24} \text{ kg}\right)}}$$
$$= 8.63 \times 10^5 \text{ s}$$

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The time for a one-way trip from Earth to the Moon is then

$$\Delta t = \frac{1}{2}T = \left(\frac{8.63\times10^5\ \text{s}}{2}\right)\left(\frac{1\ \text{day}}{8.64\times10^4\ \text{s}}\right) = \boxed{4.99\ \text{d}}$$

P13.26 The gravitational force on a small parcel of material at the star's equator supplies the necessary centripetal acceleration:

$$\frac{GM_s m}{R_s^2} = \frac{mv^2}{R_s} = mR_s\omega^2$$

so

$$\omega = \sqrt{\frac{GM_s}{R_s^3}} = \sqrt{\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left[2\left(1.99\times10^{30}\ \text{kg}\right)\right]}{\left(10.0\times10^3\ \text{m}\right)^3}}$$

$$\omega = \boxed{1.63\times10^4\ \text{rad/s}}$$

P13.27 We find the satellite's altitude from

$$\frac{GM_J}{\left(R_J + d\right)^2} = \frac{4\pi^2\left(R_J + d\right)}{T^2}$$

where d is the altitude of the satellite above Jupiter's cloud tops. Then,

$$GM_J T^2 = 4\pi^2\left(R_J + d\right)^3$$

$$\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(1.90\times10^{27}\ \text{kg}\right)\left(9.84\times3\,600\right)^2 = 4\pi^2\left(6.99\times10^7 + d\right)^3$$

which gives

$$d = \boxed{8.92\times10^7\ \text{m}} = \boxed{89\ 200\ \text{km}}\ \text{above the planet}$$

P13.28 (a) In $T^2 = 4\pi^2 a^3/GM_{\text{central}}$ we take $a = 3.84 \times 10^8$ m.

$$M_{\text{central}} = \frac{4\pi^2 a^3}{GT^2}$$

$$= \frac{4\pi^2(3.84\times10^8\ \text{m})^3}{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(27.3\times86\ 400\ \text{s})^2}$$

$$= \boxed{6.02\times10^{24}\ \text{kg}}$$

This is a little larger than 5.98×10^{24} kg.

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(b) The Earth wobbles a bit as the Moon orbits it, so both objects move nearly in circles about their center of mass, staying on opposite sides of it. The radius of the Moon's orbit is therefore a bit less than the Earth-Moon distance.

P13.29 The speed of a planet in a circular orbit is given by

$$\sum F = ma: \quad \frac{GM_{\text{sun}}m}{r^2} = \frac{mv^2}{r} \quad \rightarrow \quad v = \sqrt{\frac{GM_{\text{sun}}}{r}}$$

(a) For Mercury, the speed is

$$v_M = \sqrt{\frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(1.99\times10^{30}\ \text{kg})}{5.79\times10^{10}\ \text{m}}}$$
$$= 4.79\times10^{4}\ \text{m/s}$$

and for Pluto,

$$v_p = \sqrt{\frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(1.99\times10^{30}\ \text{kg})}{5.91\times10^{12}\ \text{m}}}$$
$$= 4.74\times10^{3}\ \text{m/s}$$

With greater speed, Mercury will eventually move farther from the Sun than Pluto.

(b) With original distances r_P and r_M perpendicular to their lines of motion, they will be equally far from the Sun at time t, where

$$\sqrt{r_P^2 + v_P^2 t^2} = \sqrt{r_M^2 + v_M^2 t^2}$$
$$r_P^2 - r_M^2 = \left(v_M^2 - v_P^2\right)t^2$$
$$t = \sqrt{\frac{\left(5.91\times10^{12}\ \text{m}\right)^2 - \left(5.79\times10^{10}\ \text{m}\right)^2}{\left(4.79\times10^{4}\ \text{m/s}\right)^2 - \left(4.74\times10^{3}\ \text{m/s}\right)^2}}$$
$$= \sqrt{\frac{3.49\times10^{25}\ \text{m}^2}{2.27\times10^{9}\ \text{m}^2/\text{s}^2}} = 1.24\times10^{8}\ \text{s} = \boxed{3.93\ \text{yr}}$$

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Section 13.5 Gravitational Potential Energy

P13.30 (a) We compute the gravitational potential energy of the satellite-Earth system from

$$U = -\frac{GM_E m}{r}$$

$$= -\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)\left(100\ \text{kg}\right)}{(6.37+2.00)\times10^{6}\ \text{m}}$$

$$= \boxed{-4.77\times10^{9}\ \text{J}}$$

(b), (c) The satellite and Earth exert forces of equal magnitude on each other, directed $\boxed{\text{downward on the satellite and upward on Earth.}}$

The magnitude of this force is

$$F = \frac{GM_E m}{r^2}$$

$$= \frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)\left(100\ \text{kg}\right)}{\left(8.37\times10^{6}\ \text{m}\right)^2}$$

$$= \boxed{569\ \text{N}}$$

P13.31 The work done by the Moon's gravitational field is equal to the negative of the change of potential energy of the meteor-Moon system:

$$W_{\text{int}} = -\Delta U = -\left(\frac{-Gm_1m_2}{r} - 0\right)$$

$$W_{\text{int}} = \frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(7.36\times10^{22}\ \text{kg})(1.00\times10^{3}\ \text{kg})}{1.74\times10^{6}\ \text{m}}$$

$$= \boxed{2.82\times10^{9}\ \text{J}}$$

***P13.32** The enery required is equal to the change in gravitational potential energy of the object-Earth system:

$$U = -G\frac{Mm}{r} \text{ and } g = \frac{GM_E}{R_E^2} \text{ so that}$$

$$\Delta U = -GMm\left(\frac{1}{3R_E} - \frac{1}{R_E}\right) = \frac{2}{3}mgR_E$$

$$\Delta U = \frac{2}{3}(1\ 000\ \text{kg})\left(9.80\ \text{m/s}^2\right)\left(6.37\times10^{6}\ \text{m}\right) = \boxed{4.17\times10^{10}\ \text{J}}$$

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P13.33 (a) The definition of density gives

$$\rho = \frac{M_S}{\frac{4}{3}\pi r_E^2} = \frac{3\left(1.99\times10^{30}\ \text{kg}\right)}{4\pi\left(6.37\times10^{6}\ \text{m}\right)^3} = \boxed{1.84\times10^{9}\ \text{kg/m}^3}$$

(b) For an object of mass m on its surface, $mg = GM_S\, m/R_E{}^2$. Thus,

$$g = \frac{GM_S}{r_E^2} = \frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(1.99\times10^{30}\ \text{kg}\right)}{\left(6.37\times10^{6}\ \text{m}\right)^2}$$

$$= \boxed{3.27\times10^{6}\ \text{m/s}^2}$$

(c) Relative to $U_g = 0$ at infinity, the potential energy of the object-star system at the surface of the white dwarf is

$$U_g = -\frac{GM_S m}{r_E}$$

$$= -\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(1.99\times10^{30}\ \text{kg}\right)\left(1.00\ \text{kg}\right)}{6.37\times10^{6}\ \text{m}}$$

$$= \boxed{-2.08\times10^{13}\ \text{J}}$$

P13.34 (a) Energy conservation of the object-Earth system from release to radius r:

$$\left(K+U_g\right)_{\text{altitude } h} = \left(K+U_g\right)_{\text{radius } r}$$

$$0 - \frac{GM_E m}{R_E + h} = \frac{1}{2}mv^2 - \frac{GM_E m}{r}$$

$$v = \left[2GM_E\left(\frac{1}{r} - \frac{1}{R_E + h}\right)\right]^{1/2} = -\frac{dr}{dt}$$

(b) $\displaystyle\int_i^f dt = \int_i^f -\frac{dr}{v} = \int_f^i \frac{dr}{v}$. The time of fall is, suppressing units,

$$\Delta t = \int_{R_E}^{R_E+h}\left[2GM_E\left(\frac{1}{r} - \frac{1}{R_E+h}\right)\right]^{-1/2} dr$$

$$\Delta t = \left(2\times6.67\times10^{-11}\times5.98\times10^{24}\right)^{-1/2} \times \int_{6.37\times10^{6}\ \text{m}}^{6.87\times10^{6}\ \text{m}}\left[\left(\frac{1}{r} - \frac{1}{6.87\times10^{6}\ \text{m}}\right)\right]^{-1/2} dr$$

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We can enter this expression directly into a mathematical calculation program.

Alternatively, to save typing we can change variables to $u = \frac{r}{10^6}$.

Then

$$\Delta t = \left(7.977 \times 10^{14}\right)^{-1/2} \int_{6.37}^{6.87} \left(\frac{1}{10^6 u} - \frac{1}{6.87 \times 10^6} \right)^{-1/2} 10^6 \, du$$

$$= 3.541 \times 10^{-8} \frac{10^6}{\left(10^6\right)^{-1/2}} \int_{6.37}^{6.87} \left(\frac{1}{u} - \frac{1}{6.87} \right)^{-1/2} du$$

A mathematics program returns the value 9.596 for this integral, giving for the time of fall

$$\Delta t = 3.541 \times 10^{-8} \times 10^9 \times 9.596 = 339.8 = \boxed{340 \text{ s}}$$

P13.35 (a) Since the particles are located at the corners of an equilateral triangle, the distances between all particle pairs is equal to 0.300 m. The gravitational potential energy of the system is then

$$U_{\text{Tot}} = U_{12} + U_{13} + U_{23} = 3U_{12} = 3\left(-\frac{Gm_1 m_2}{r_{12}} \right)$$

$$U_{\text{Tot}} = -\frac{3\left(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2\right)\left(5.00 \times 10^{-3} \text{ kg}\right)^2}{0.300 \text{ m}}$$

$$= \boxed{-1.67 \times 10^{-14} \text{ J}}$$

(b) Each particle feels a net force of attraction toward the midpoint between the other two. Each moves toward the center of the triangle with the same acceleration. They collide simultaneously at the center of the triangle.

Section 13.6 Energy Considerations in Planetary and Satellite Motion

P13.36 We use the isolated system model for energy:

$$K_i + U_i = K_f + U_f$$

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$$\frac{1}{2}mv_i^2 + GM_E m\left(\frac{1}{r_f} - \frac{1}{r_i}\right) = \frac{1}{2}mv_f^2$$

which becomes

$$\frac{1}{2}v_i^2 + GM_E\left(0 - \frac{1}{R_E}\right) = \frac{1}{2}v_f^2$$

or $$v_f^2 = v_1^2 - \frac{2GM_E}{R_E}$$

and $$v_f = \left(v_1^2 - \frac{2GM_E}{R_E}\right)^{1/2}$$

$$v_f = \left\{\left(2.00\times10^4\ \text{m/s}\right)^2 - \left[\frac{2\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)}{6.37\times10^6\ \text{m}}\right]\right\}^{1/2}$$

$$= \boxed{1.66\times10^4\ \text{m/s}}$$

***P13.37** To determine the energy transformed to internal energy, we begin by calculating the change in kinetic energy of the satellite. To find the initial kinetic energy, we use

$$\frac{v_i^2}{R_E + h} = \frac{GM_E}{(R_E + h)^2}$$

which gives

$$K_i = \frac{1}{2}mv_i^2 = \frac{1}{2}\left(\frac{GM_E m}{R_E + h}\right)$$

$$= \frac{1}{2}\left[\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)(500\ \text{kg})}{6.37\times10^6\ \text{m} + 0.500\times10^6\ \text{m}}\right]$$

$$= 1.45\times10^{10}\ \text{J}$$

Also, $$K_f = \frac{1}{2}mv_f^2 = \frac{1}{2}(500\ \text{kg})\left(2.00\times10^3\ \text{m/s}\right)^2 = 1.00\times10^9\ \text{J}.$$

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The change in gravitational potential energy of the satellite-Earth system is

$$\Delta U = \frac{GM_E m}{R_i} - \frac{GM_E m}{R_f} = GM_E m\left(\frac{1}{R_i} - \frac{1}{R_f}\right)$$

$$= \left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)(500\ \text{kg})$$

$$\times\left(-1.14\times10^{-8}\ \text{m}^{-1}\right)$$

$$= -2.27\times10^{9}\ \text{J}$$

The energy transformed into internal energy due to friction is then

$$\Delta E_{\text{int}} = K_i - K_f - \Delta U = (14.5 - 1.00 + 2.27)\times10^{9}\ \text{J}$$

$$= \boxed{1.58\times10^{10}\ \text{J}}$$

P13.38 To obtain the orbital velocity, we use

$$\sum F = \frac{mMG}{R^2} = \frac{mv^2}{R}$$

or $$v = \sqrt{\frac{MG}{R}}$$

We can obtain the escape velocity from

$$\frac{1}{2}mv_{\text{esc}}^2 = \frac{mMG}{R}$$

or $$v_{\text{esc}} = \sqrt{\frac{2MG}{R}} = \boxed{\sqrt{2}v}$$

P13.39 (a) The total energy of the satellite-Earth system at a given orbital altitude is given by

$$E_{\text{tot}} = -\frac{GMm}{2r}$$

The energy needed to increase the satellite's orbit is then, suppressing units,

$$\Delta E = \frac{GMm}{2}\left(\frac{1}{r_i} - \frac{1}{r_f}\right)$$

$$= \frac{\left(6.67\times10^{-11}\right)\left(5.98\times10^{24}\right)}{2}\frac{10^3\ \text{kg}}{10^3\ \text{m}}\left(\frac{1}{6\,370+100} - \frac{1}{6\,370+200}\right)$$

$$\Delta E = 4.69\times10^{8}\ \text{J} = \boxed{469\ \text{MJ}}$$

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(b) Both in the original orbit and in the final orbit, the total energy is negative, with an absolute value equal to the positive kinetic energy. The potential energy is negative and twice as large as the total energy. As the satellite is lifted from the lower to the higher orbit, the gravitational energy increases, the kinetic energy decreases, and the total energy increases. The value of each becomes closer to zero. Numerically, the gravitational energy increases by 938 MJ, the kinetic energy decreases by 469 MJ, and the total energy increases by 469 MJ.

P13.40 (a) The major axis of the orbit is $2a = 50.5$ AU so $a = 25.25$ AU.

Further, in the textbook's diagram of an ellipse, $a + c = 50$ AU, so $c = 24.75$ AU. Then

$$e = \frac{c}{a} = \frac{24.75}{25.25} = \boxed{0.980}$$

(b) In $T^2 = K_s a^3$ for objects in solar orbit, the Earth gives us

$$(1\text{ yr})^2 = K_s (1\text{ AU})^3 \quad K_s = \frac{(1\text{ yr})^2}{(1\text{ AU})^3}$$

Then

$$T^2 = \frac{(1\text{ yr})^2}{(1\text{ AU})^3}(25.25\text{ AU})^3 \quad \rightarrow \quad T = \boxed{127\text{ yr}}$$

(c) $$U = -\frac{GMm}{r}$$

$$= -\frac{\left(6.67\times10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2\right)\left(1.99\times10^{30}\text{ kg}\right)\left(1.20\times10^{10}\text{ kg}\right)}{50\left(1.496\times10^{11}\text{ m}\right)}$$

$$= \boxed{-2.13\times10^{17}\text{ J}}$$

***P13.41** For her jump on Earth,

$$\frac{1}{2}mv_i^2 = mgy_f \qquad \textbf{[1]}$$

which gives

$$v_i = \sqrt{2gy_f} = \sqrt{2(9.80\text{ m/s})(0.500\text{ m})} = 3.13\text{ m/s}$$

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We assume that she has the same takeoff speed on the asteroid. Here

$$\frac{1}{2}mv_i^2 - \frac{GM_A m}{R_A} = 0 + 0 \qquad \textbf{[2]}$$

The equality of densities between planet and asteroid,

$$\rho = \frac{M_E}{\frac{4}{3}\pi R_E^3} = \frac{M_A}{\frac{4}{3}\pi R_A^3}$$

implies

$$M_A = \left(\frac{R_A}{R_E}\right)^3 M_E \qquad \textbf{[3]}$$

Note also at Earth's surface

$$g = \frac{GM_E}{R_E^2} \qquad \textbf{[4]}$$

Combining the equations [2], [1], [3], and [4] by substitution gives

$$\frac{1}{2}v_i^2 = \frac{GM_A}{R_A}$$

$$\frac{GM_E}{R_E^2}y_f = \frac{GM_E R_A^2}{R_E^3}$$

$$R_A^2 = y_f R_E = (0.500\text{ m})\left(6.37\times10^6\text{ m}\right)$$

$$R_A = \boxed{1.78\times10^3\text{ m}}$$

***P13.42** For a satellite in an orbit of radius r around the Earth, the total energy of the satellite-Earth system is $E = -\frac{GM_E}{2r}$. Thus, in changing from a circular orbit of radius $r = 2R_E$ to one of radius $r = 3R_E$, the required work is

$$W = \Delta E = -\frac{GM_E m}{2r_f} + \frac{GM_E m}{2r_i} = GM_E m\left[\frac{1}{4R_E} - \frac{1}{6R_E}\right] = \boxed{\frac{GM_E m}{12R_E}}$$

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***P13.43** (a) The work must provide the increase in gravitational energy:

$$W = \Delta U_g = U_{gf} - U_{gi}$$

$$= -\frac{GM_E M_p}{r_f} + \frac{GM_E M_p}{r_i}$$

$$= -\frac{GM_E M_p}{R_E + y} + \frac{GM_E M_p}{R_E}$$

$$= GM_E M_p \left(\frac{1}{R_E} - \frac{1}{R_E + y} \right)$$

$$= \left(\frac{6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2}{\text{kg}^2} \right) (5.98 \times 10^{24}\ \text{kg})(100\ \text{kg}) \times \left(\frac{1}{6.37 \times 10^6\ \text{m}} - \frac{1}{7.37 \times 10^6\ \text{m}} \right)$$

$$W = \boxed{850\ \text{MJ}}$$

(b) In a circular orbit, gravity supplies the centripetal force:

$$\frac{GM_E M_p}{(R_E + y)^2} = \frac{M_p v^2}{R_E + y}$$

Then,

$$\frac{1}{2} M_p v^2 = \frac{1}{2} \frac{GM_E M_p}{(R_E + y)}$$

So, additional work = kinetic energy required is

$$\Delta W = \frac{1}{2} \frac{(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2 / \text{kg}^2)(5.98 \times 10^{24}\ \text{kg})(100\ \text{kg})}{7.37 \times 10^6\ \text{m}}$$

$$= \boxed{2.71 \times 10^9\ \text{J}}$$

P13.44 (a) The escape velocity from the solar system, starting at Earth's orbit, is given by

$$v_{\text{solar escape}} = \sqrt{\frac{2M_{\text{Sun}} G}{R_{\text{Sun}}}}$$

$$= \sqrt{\frac{2(1.99 \times 10^{30}\ \text{kg})(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2 / \text{kg}^2)}{1.50 \times 10^9\ \text{m}}}$$

$$= \boxed{42.1\ \text{km/s}}$$

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(b) Let x represent the variable distance from the Sun. Then,

$$v = \sqrt{\frac{2M_{\text{Sun}}G}{x}} \quad \rightarrow \quad x = \frac{v^2}{2M_{\text{Sun}}G}$$

If $v = \dfrac{125\,000 \text{ km}}{3\,600 \text{ s}} = 34.7 \text{ m/s}$, then

$$x = \frac{v^2}{2M_{\text{Sun}}G} = \frac{(34.7 \text{ m/s})^2}{2\left(1.99\times10^{30} \text{ kg}\right)\left(6.67\times10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2\right)}$$

$$= \boxed{2.20\times10^{11} \text{ m}}$$

Note that at or beyond the orbit of Mars, 125 000 km/h is sufficient for escape.

P13.45 $F_c = F_G$ gives $\dfrac{mv^2}{r} = \dfrac{GmM_E}{r^2}$

which reduces to $v = \sqrt{\dfrac{GM_E}{r}}$

and period $= \dfrac{2\pi r}{v} = 2\pi r\sqrt{\dfrac{r}{GM_E}}$.

(a) $r = R_E + 200 \text{ km} = 6\,370 \text{ km} + 200 \text{ km} = 6\,570 \text{ km}$

Thus,

$$\text{period} = 2\pi\left(6.57\times10^6 \text{ m}\right) \times\sqrt{\frac{6.57\times10^6 \text{ m}}{\left(6.67\times10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24} \text{ kg}\right)}}$$

$$T = 5.30\times10^3 \text{ s} = 88.3 \text{ min} = \boxed{1.47 \text{ h}}$$

(b) $v = \sqrt{\dfrac{GM_E}{r}}$

$$= \sqrt{\frac{\left(6.67\times10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24} \text{ kg}\right)}{6.57\times10^6 \text{ m}}}$$

$$= \boxed{7.79 \text{ km/s}}$$

(c) $K_f + U_f = K_i + U_i +$ energy input gives

$$\text{input} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 + \left(\frac{-GM_Em}{r_f}\right) - \left(\frac{-GM_Em}{r_i}\right) \qquad \textbf{[1]}$$

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$$r_i = R_E = 6.37 \times 10^6 \text{ m}$$

$$v_i = \frac{2\pi R_E}{86\,400 \text{ s}} = 4.63 \times 10^2 \text{ m/s}$$

Substituting the appropriate values into [1] yields:

$$\text{minimum energy input} = \boxed{6.43 \times 10^9 \text{ J}}$$

P13.46 The gravitational force supplies the needed centripetal acceleration.

Thus, $$\frac{GM_E m}{(R_E + h)^2} = \frac{mv^2}{R_E + h} \qquad \text{or} \qquad v^2 = \frac{GM_E}{R_E + h}$$

(a) $$T = \frac{2\pi r}{v} = \frac{2\pi (R_E + h)}{\sqrt{\frac{GM_E}{(R_E + h)}}} = \boxed{2\pi \sqrt{\frac{(R_E + h)^3}{GM_E}}}$$

(b) $$v = \boxed{\sqrt{\frac{GM_E}{R_E + h}}}$$

(c) Minimum energy input is

$$\Delta E_{\min} = (K_f + U_{gf}) - (K_i - U_{gi})$$

This choice has the object starting with energy

$$K_i = \frac{1}{2} m v_i^2$$

with $v_i = \dfrac{2\pi R_E}{1.00 \text{ day}} = \dfrac{2\pi R_E}{86\,400 \text{ s}}$ and $U_{gi} = -\dfrac{GM_E m}{R_E}$.

Thus,

$$\Delta E_{\min} = \frac{1}{2} m \left(\frac{GM_E}{R_E + h} \right) - \frac{GM_E m}{R_E + h} - \frac{1}{2} m \left[\frac{4\pi^2 R_E^2}{(86\,400 \text{ s})^2} \right] + \frac{GM_E m}{R_E}$$

or $$\Delta E_{\min} = \boxed{GM_E m \left[\frac{R_E + 2h}{2R_E (R_E + h)} \right] - \frac{2\pi^2 R_E^2 m}{(86\,400 \text{ s})^2}}.$$

P13.47 (a) Gravitational screening does not exist. The presence of the satellite has no effect on the force the planet exerts on the rocket.

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(b) The rocket has a gravitational potential energy with respect to Ganymede

$$U_1 = -\frac{Gm_1m_2}{r} = -\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2\right)m_2\left(1.495\times10^{23}\ \text{kg}\right)}{\left(2.64\times10^{6}\ \text{m}\right)\text{kg}^2}$$

$$U_1 = \left(-3.78\times10^{6}\ \text{m}^2/\text{s}^2\right)m_2$$

The rocket's gravitational potential energy with respect to Jupiter at the distance of Ganymede is

$$U_2 = -\frac{Gm_1m_2}{r} = -\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2\right)m_2\left(1.90\times10^{27}\ \text{kg}\right)}{\left(1.071\times10^{9}\ \text{m}\right)\text{kg}^2}$$

$$U_2 = \left(-1.18\times10^{8}\ \text{m}^2/\text{s}^2\right)m_2$$

To escape from both requires

$$\frac{1}{2}m_2v_{\text{esc}}^2 = +\left[\left(3.78\times10^{6}+1.18\times10^{8}\right)\text{m}^2/\text{s}^2\right]m_2$$

$$v_{\text{esc}} = \sqrt{2\times\left(1.22\times10^{8}\ \text{m}^2/\text{s}^2\right)} = \boxed{15.6\ \text{km/s}}$$

P13.48 (a) For the satellite $\sum F = ma;\ \frac{GmM_E}{r^2} = \frac{mv_i^2}{r}$ gives

$$\boxed{v_i = \left(\frac{GM_E}{r}\right)^{1/2}}$$

(b) Conservation of momentum in the forward direction for the exploding satellite gives:

$$\left(\sum mv\right)_i = \left(\sum mv\right)_f$$

$$5mv_i = 4mv + m0$$

$$v = \frac{5}{4}v_i = \boxed{\frac{5}{4}\left(\frac{GM_E}{r}\right)^{1/2}}$$

(c) With velocity perpendicular to radius, the orbiting fragment is at perigee. Its apogee distance and speed are related to r and v by $4mrv = 4mr_fv_f$ and

$$\frac{1}{2}4mv^2 - \frac{GM_E4m}{r} = \frac{1}{2}4mv_f^2 - \frac{GM_E4m}{r_f}$$

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Substituting $v_f = \frac{vr}{r_f}$ we have

$$\frac{1}{2}v^2 - \frac{GM_E}{r} = \frac{1}{2}\frac{v^2 r^2}{r_f^2} - \frac{GM_E}{r_f}$$

Further, substituting $v^2 = \frac{25}{16}\frac{GM_E}{r}$ gives

$$\frac{25}{32}\frac{GM_E}{r} - \frac{GM_E}{r} = \frac{25}{32}\frac{GM_E r}{r_f^2} - \frac{GM_E}{r_f}$$

$$\frac{-7}{32r} = \frac{25r}{32r_f^2} - \frac{1}{r_f}$$

Clearing fractions we have $-7r_f^2 = 25r^2 - 32rr_f$, or

$$7\left(\frac{r_f}{r}\right)^2 - 32\left(\frac{r_f}{r}\right) + 25 = 0$$

giving $\frac{r_f}{r} = \frac{+32 \pm \sqrt{32^2 - 4(7)(25)}}{14} = \frac{50}{14}$ or $\frac{14}{14}$.

The latter root describes the starting point.

The outer end of the orbit has $\frac{r_f}{r} = \frac{25}{7}$: $\boxed{r_f = \frac{25r}{7}}$

***P13.49** The height attained is not small compared to the radius of the Earth, so $U = mgy$ does not apply; $U = -\frac{GM_1M_2}{r}$ does. From launch to apogee at height h, conservation of energy gives

$$K_i + U_i + \Delta E_{\text{mech}} = K_f + U_f$$

$$\frac{1}{2}M_p v_i^2 - \frac{GM_E M_p}{R_E} + 0 = 0 - \frac{GM_E M_p}{R_E + h}$$

The mass of the projectile cancels out, giving

$$\frac{1}{2}v_i^2 - \frac{GM_E}{R_E} = \frac{GM_E}{R_E + h}$$

$$R_E + h = \frac{GM_E}{\frac{1}{2}v_i^2 - \frac{GM_E}{R_E}}$$

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$$h = \frac{GM_E}{\frac{1}{2}v_i^2 - \frac{GM_E}{R_E}} - R_E$$

$$= \frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\ \text{kg})}{\frac{1}{2}(10.0\times10^{3}\ \text{m/s})^2 - \frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\ \text{kg})}{(6.37\times10^{6}\ \text{m})}} - 6.37\times10^{6}\ \text{m}$$

$$= \boxed{2.52\times10^{7}\ \text{m}}$$

Additional Problems

P13.50 (a) When the rocket engine shuts off at an altitude of 250 km, we may consider the rocket to be beyond Earth's atmosphere. Then, its mechanical energy will remain constant from that instant until it comes to rest momentarily at the maximum altitude. That is, $KE_f + PE_f = KE_i + PE_i$, or

$$0 - \frac{GM_E \cancel{m}}{r_{\max}} = \frac{1}{2}\cancel{m}v_i^2 - \frac{GM_E \cancel{m}}{r_i} \qquad \text{or} \qquad \frac{1}{r_{\max}} = -\frac{v_i^2}{2GM_E} + \frac{1}{r_i}$$

With $r_l = R_E + 250\ \text{km} = 6.37\times10^6\ \text{m} + 250\times10^3\ \text{m} = 6.62\times10^6\ \text{m}$ and $v_i = 6.00\ \text{km/s} = 6.00\times10^3\ \text{m/s}$, this gives

$$\frac{1}{r_{\max}} = -\frac{(6.00\times10^{3}\ \text{m/s})^2}{2(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\ \text{kg})} + \frac{1}{6.62\times10^{6}\ \text{m}}$$

$$= 1.06\times10^{-7}\ \text{m}^{-1}$$

or $r_{\max} = 9.44\times10^6\ \text{m}$. The maximum distance from Earth's surface is then

$$h_{max} = r_{max} - R_E = 9.44\times10^6\ \text{m} - 6.37\times10^6\ \text{m} = \boxed{3.07\times10^6\text{m}}$$

(b) If the rocket were fired from a launch site on the equator, it would have a significant eastward component of velocity because of the Earth's rotation about its axis. Hence, compared to being fired from the South Pole, the rocket's initial speed would be greater, and the rocket would travel farther from Earth.

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P13.51 For a 6.00-km diameter cylinder, $r = 3\,000$ m and to simulate $1g = 9.80\ \text{m/s}^2$,

$$g = \frac{v^2}{r} = \omega^2 r$$

$$\omega = \sqrt{\frac{g}{r}} = \boxed{0.057\ 2\ \text{rad/s}}$$

The required rotation rate of the cylinder is $\boxed{\frac{1\ \text{rev}}{110\ \text{s}}}$.

(For a description of proposed cities in space, see Gerard K. O'Neill in *Physics Today*, Sept. 1974. and the Wikipedia article on "Rotating Wheel Space Station at http://en.wikipedia.org/wiki/Rotating_wheel_space_station)

***P13.52** To approximate the height of the sulfur, set $\frac{mv^2}{2} = mg_{Io}h,$ with h = 70 000 m and $g_{Io} = \frac{GM}{r^2} = 1.79\ \text{m/s}^2$. This gives

$$v = \sqrt{2g_{Io}h} = \sqrt{2(1.79\ \text{m/s}^2)(70\,000\ \text{m})}$$
$$\approx 500\ \text{m/s}\ \ (\text{over } 1\,000\ \text{mi/h})$$

We can obtain a more precise answer from conservation of energy:

$$\frac{1}{2}mv^2 - \frac{GMm}{r_1} = -\frac{GMm}{r_2}$$

$$\frac{1}{2}v^2 = (6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(8.90\times10^{22}\ \text{kg}) \times\left(\frac{1}{1.82\times10^6\ \text{m}} - \frac{1}{1.89\times10^6\ \text{m}}\right)$$

$$v = \boxed{492\ \text{m/s}}$$

***P13.53** (a) The radius of the satellite's orbit is

$$r = R_E + h = 6.37\times10^6\ \text{m} + 2.80\times10^6\ \text{m} = 9.17\times10^6\ \text{m}$$

Then, modifying Kepler's third law for orbital motion about the Earth rather than the Sun, we have

$$T^2 = \left(\frac{4\pi^2}{GM_E}\right)r^3 = \frac{4\pi^2(9.17\times10^6\ \text{m})^3}{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\ \text{kg})}$$
$$= 7.63\times10^7\ \text{s}^2$$

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or $$T = \left(8.74 \times 10^3 \text{ s}\right)\left(\frac{1 \text{ h}}{3\ 600 \text{ s}}\right) = \boxed{2.43 \text{ h}}$$

(b) The constant tangential speed of the satellite is

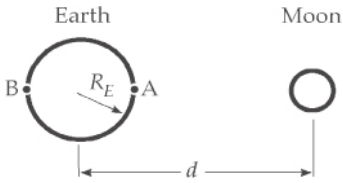
$$v = \frac{2\pi r}{T} = \frac{2\pi\left(9.17 \times 10^6 \text{ m}\right)}{8.74 \times 10^3 \text{ s}} = 6.60 \times 10^3 \text{ m/s} = \boxed{6.60 \text{ km/s}}$$

(c) The satellite's only acceleration is centripetal acceleration, so

$$a = a_c = \frac{v^2}{r} = \frac{\left(6.60 \times 10^3 \text{ m/s}\right)^2}{9.17 \times 10^6 \text{ m}} = \boxed{4.74 \text{ m/s}^2 \text{ toward the Earth}}$$

P13.54 If one uses the result $v = \sqrt{\frac{GM}{r}}$ and the relation $v = (2\pi r/T)$, one finds the radius of the orbit to be smaller than the radius of the Earth, so the spacecraft would need to be in orbit underground.

P13.55 The acceleration of an object at the center of the Earth due to the gravitational force of the Moon is given by $a = G\frac{M_M}{d^2}$.



ANS. FIG. P13.55

At the point A nearest the Moon,

$$a_+ = G\frac{M_M}{\left(d - R_E\right)^2}$$

At the point B farthest from the Moon,

$$a_- = G\frac{M_M}{\left(d + R_E\right)^2}$$

From the above, we have

$$\frac{\Delta g_M}{g} = \frac{\left(a_+ - a_-\right)}{g} = \frac{GM_M}{g}\left[\frac{1}{\left(d - R_E\right)^2} - \frac{1}{\left(d + R_E\right)^2}\right]$$

Evaluating this expression, we find across the planet

$$\frac{\Delta g_M}{g} = \frac{\left(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2\right)\left(7.36 \times 10^{22} \text{ kg}\right)}{9.80 \text{ m/s}^2} \times \left[\frac{1}{\left(3.84 \times 10^8 \text{ m} - 6.37 \times 10^6 \text{ m}\right)^2} - \frac{1}{\left(3.84 \times 10^8 \text{ m} + 6.37 \times 10^6 \text{ m}\right)^2}\right]$$

$$= \boxed{2.25 \times 10^{-7}}$$

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P13.56 (a) The only force acting on the astronaut is the normal force exerted on him by the "floor" of the cabin. The normal force supplies the centripetal force:

$$F_c = \frac{mv^2}{r} \qquad \text{and} \qquad n = \frac{mg}{2}$$

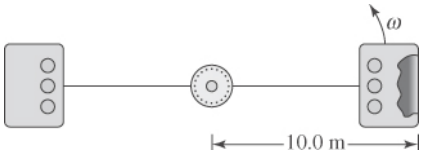
This gives

$$\frac{mv^2}{r} = \frac{mg}{2} \rightarrow v = \sqrt{\frac{gr}{2}}$$

$$v = \sqrt{\frac{(9.80\,\text{m/s}^2)(10.0\,\text{m})}{2}} \rightarrow v = 7.00\ \text{m/s}$$

Since $v = r\omega$, we have

$$\omega = \frac{v}{r} = \frac{7.00\ \text{m/s}}{10.0\ \text{m}} = \boxed{0.700\ \text{rad/s}}$$



ANS. FIG. P13.56

(b) Because his feet stay in place on the floor, his head will be moving at the same tangential speed as his feet. However, his feet and his head are travelling in circles of different radii.

(c) If he stands up without holding on to anything with his hands, the only force on his body is radial. Because the wall of the cabin near the traveler's head moves in a smaller circle, it moves at a slower tangential speed than that of the traveler's head so his head moves toward the wall—if he is not careful, there could be a collision. This is an example of the Coriolis force investigated in Section 6.3. Holding onto to a rigid support with his hands will provide a tangential force to the traveler to slow the upper part of his body down.

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P13.57 (a) Ignoring air resistance, the energy conservation for the object-Earth system from firing to apex is given by,

$$\left(K+U_g\right)_i=\left(K+U_g\right)_f$$

$$\frac{1}{2}mv_i^2-\frac{GmM_E}{R_E}=0-\frac{GmM_E}{R_E+h}$$

where $\frac{1}{2}mv_{\text{esc}}^2=\frac{GmM_E}{R_E}$. Then

$$\frac{1}{2}v_i^2-\frac{1}{2}v_{\text{esc}}^2=-\frac{1}{2}v_{\text{esc}}^2\frac{R_E}{R_E+h}$$

$$v_{\text{esc}}^2-v_i^2=\frac{v_{\text{esc}}^2R_E}{R_E+h}$$

$$\frac{1}{v_{\text{esc}}^2-v_i^2}=\frac{R_E+h}{v_{\text{esc}}^2R_E}$$

$$h=\frac{v_{\text{esc}}^2R_E}{v_{\text{esc}}^2-v_i^2}-R_E=\frac{v_{\text{esc}}^2R_E-v_{\text{esc}}^2R_E+v_i^2R_E}{v_{\text{esc}}^2-v_i^2}$$

$$h=\frac{R_Ev_i^2}{v_{\text{esc}}^2-v_i^2}$$

$$h=\frac{\left(6.37\times10^6\text{ m}\right)\left(8.76\text{ km/s}\right)^2}{\left(11.2\text{ km/s}\right)^2-\left(8.76\text{ km/s}\right)^2}=\boxed{1.00\times10^7\text{ m}}$$

(b) The fall of the meteorite is the time-reversal of the upward flight of the projectile, so it is described by the same energy equation:

$$v_i^2=v_{\text{esc}}^2\left(1-\frac{R_E}{R_E+h}\right)$$

$$=v_{\text{esc}}^2\left(\frac{h}{R_E+h}\right)$$

$$=\left(11.2\times10^3\text{ m/s}\right)^2\left(\frac{2.51\times10^7\text{ m}}{6.37\times10^6\text{ m}+2.51\times10^7\text{ m}}\right)$$

$$=1.00\times10^8\text{ m}^2/\text{s}^2$$

$$v_i=\boxed{1.00\times10^4\text{ m/s}}$$

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P13.58 (a) Ignoring air resistance, the energy conservation for the object-Earth system from firing to apex is given by,

$$\left(K+U_g\right)_i=\left(K+U_g\right)_f$$

$$\frac{1}{2}mv_i^2-\frac{GmM_E}{R_E}=0-\frac{GmM_E}{R_E+h}$$

where $\frac{1}{2}mv_{\text{esc}}^2=\frac{GmM_E}{R_E}$. Then

$$\frac{1}{2}v_i^2-\frac{1}{2}v_{\text{esc}}^2=-\frac{1}{2}v_{\text{esc}}^2\frac{R_E}{R_E+h}$$

$$v_{\text{esc}}^2-v_i^2=\frac{v_{\text{esc}}^2R_E}{R_E+h}$$

$$\frac{1}{v_{\text{esc}}^2-v_i^2}=\frac{R_E+h}{v_{\text{esc}}^2R_E}$$

$$h=\frac{v_{\text{esc}}^2R_E}{v_{\text{esc}}^2-v_i^2}-R_E=\frac{v_{\text{esc}}^2R_E-v_{\text{esc}}^2R_E+v_i^2R_E}{v_{\text{esc}}^2-v_i^2}$$

$$\boxed{h=\frac{R_Ev_i^2}{v_{\text{esc}}^2-v_i^2}}$$

(b) The fall of the meteorite is the time-reversal of the upward flight of the projectile, so it is described by the same energy equation. From (a) above, replacing v_i with v_f, we have

$$v_f^2=v_{\text{esc}}^2-v_{\text{esc}}^2\frac{R_E}{R_E+h}$$

$$v_f^2=v_{\text{esc}}^2\left(1-\frac{R_E}{R_E+h}\right)$$

$$\boxed{v_f=v_{\text{esc}}\sqrt{\frac{h}{R_E+h}}}$$

(c) With $v_i << v_{esc}$, $h\approx\frac{R_Ev_i^2}{v_{\text{esc}}^2}=\frac{R_Ev_i^2R_E}{2GM_E}$. But $g=\frac{GM_E}{R_E^2}$, so $h=\frac{v_i^2}{2g}$, in agreement with $0^2=v_i^2+2\left(-g\right)\left(h-0\right)$.

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P13.59 (a) Let R represent the radius of the asteroid. Then its volume is $\frac{4}{3}\pi R^3$ and its mass is $\rho\frac{4}{3}\pi R^3$. For your orbital motion, $\sum F = ma$ gives

$$\frac{Gm_1m_2}{R^2} = \frac{m_2v^2}{R} \quad \rightarrow \quad \frac{G\rho 4\pi R^3}{3R^2} = \frac{v^2}{R}$$

solving for R,

$$R = \left(\frac{3v^2}{G\rho 4\pi}\right)^{1/2}$$

$$= \left[\frac{3(8.50\ \text{m/s})^2}{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(1\,100\ \text{kg/m}^3)4\pi}\right]^{1/2}$$

$$= \boxed{1.53\times10^4\ \text{m}}$$

(b) $$\rho\frac{4}{3}\pi R^3 = (1\,100\ \text{kg/m}^3)\frac{4}{3}\pi(1.53\times10^4\ \text{m})^3 = \boxed{1.66\times10^{16}\ \text{kg}}$$

(c) $$v = \frac{2\pi R}{T} \quad T = \frac{2\pi R}{v} = \frac{2\pi(1.53\times10^4\ \text{m})}{8.5\ \text{m/s}} = \boxed{1.13\times10^4\ \text{s}} = 3.15\ \text{h}$$

(d) For an illustrative model, we take your mass as 90.0 kg and assume the asteroid is originally at rest. Angular momentum is conserved for the asteroid-you system:

$$\sum L_i = \sum L_f$$

$$0 = m_2vR - I\omega$$

$$0 = m_2vR - \frac{2}{5}m_1R^2\frac{2\pi}{T_{\text{asteroid}}}$$

$$m_2v = \frac{4\pi}{5}\frac{m_1R}{T_{\text{asteroid}}}$$

$$T_{\text{asteroid}} = \frac{4\pi m_1R}{5m_2v} = \frac{4\pi(1.66\times10^{16}\ \text{kg})(1.53\times10^4\ \text{m})}{5(90.0\ \text{kg})(8.50\ \text{m/s})}$$

$$= 8.37\times10^{17}\ \text{s} = 26.5\ \text{billion years}$$

Thus your running does not produce significant rotation of the asteroid if it is originally stationary and does not significantly affect any rotation it does have.

This problem is realistic. Many asteroids, such as Ida and Eros, are roughly 30 km in diameter. They are typically irregular in

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shape and not spherical. Satellites such as Phobos (of Mars), Adrastea (of Jupiter), Calypso (of Saturn), and Ophelia (of Uranus) would allow a visitor the same experience of easy orbital motion. So would many Kuiper Belt objects.

P13.60 (a) The two appropriate isolated system models are conservation of momentum and conservation of energy applied to the system consisting of the two spheres.

(b) Applying conservation of momentum to the system, we find

$$m_1\vec{\mathbf{v}}_{1i} + m_2\vec{\mathbf{v}}_{2i} = m_1\vec{\mathbf{v}}_{1f} + m_2\vec{\mathbf{v}}_{2f}$$

$$0 + 0 = M\vec{\mathbf{v}}_{1f} + 2M\vec{\mathbf{v}}_{2f}$$

$$\vec{\mathbf{v}}_{1f} = \boxed{-2\vec{\mathbf{v}}_{2f}}$$

(c) Applying conservation of energy to the system, we find

$$K_i + U_i + \Delta E = K_f + U_f$$

$$0 - \frac{Gm_1m_2}{r_i} + 0 = \frac{1}{2}m_1v_{1f}^2 + \frac{1}{2}m_2v_{2f}^2 - \frac{Gm_1m_2}{r_f}$$

$$-\frac{GM(2M)}{12R} = \frac{1}{2}Mv_{1f}^2 + \frac{1}{2}(2M)v_{2f}^2 - \frac{GM(2M)}{4R}$$

$$\frac{1}{2}Mv_{1f}^2 = \frac{GM}{2R} - \frac{GM}{6R} - v_{2f}^2$$

$$v_{1f} = \boxed{\sqrt{\frac{2GM}{3R} - 2v_{2f}^2}}$$

(d) Combining the results for parts (b) and (c),

$$2v_{2f} = \sqrt{\frac{2GM}{3R} - 2v_{2f}^2}$$

$$6v_{2f}^2 = \frac{2GM}{3R}$$

$$v_2 = \boxed{\frac{1}{3}\sqrt{G\frac{M}{R}}} \qquad v_1 = \boxed{\frac{2}{3}\sqrt{G\frac{M}{R}}}$$

P13.61 (a) At infinite separation $U = 0$ and at rest $K = 0$. Since the system is isolated, the energy and momentum of the two-planet system is conserved. We have

$$0 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{Gm_1m_2}{d} \qquad \textbf{[1]}$$

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and

$$0 = m_1 v_1 - m_2 v_2 \qquad \textbf{[2]}$$

because the initial momentum of the system is zero.

Combine equations [1] and [2]:

$$\boxed{v_1 = m_2\sqrt{\frac{2G}{d\left(m_1 + m_2\right)}}} \qquad \text{and} \qquad \boxed{v_2 = m_1\sqrt{\frac{2G}{d\left(m_1 + m_2\right)}}}$$

The relative velocity is then

$$v_r = v_1 - \left(-v_2\right) = \boxed{\sqrt{\frac{2G\left(m_1 + m_2\right)}{d}}}$$

(b) The instant before the collision, the distance between the planets is $d = r_1 + r_2$. Substitute given numerical values into the equation found for v_1 and v_2 in part (a) to find

$$v_1 = 1.03 \times 10^4 \text{ m/s} \qquad \text{and} \qquad v_2 = 2.58 \times 10^3 \text{ m/s}$$

Therefore,

$$K_1 = \frac{1}{2} m_1 v_1^2 = \boxed{1.07 \times 10^{32} \text{ J}} \text{ and } K_2 = \frac{1}{2} m_2 v_2^2 = \boxed{2.67 \times 10^{31} \text{ J}}$$

P13.62 (a) The free-fall acceleration produced by the Earth is

$$g = \frac{GM_E}{r^2} = GM_E r^{-2} \text{ (directed downward)}$$

Its rate of change is

$$\frac{dg}{dr} = GM_E\left(-2\right)r^{-3} = -2GM_E r^{-3}$$

The minus sign indicates that g decreases with increasing height. At the Earth's surface,

$$\boxed{\frac{dg}{dr} = -\frac{2GM_E}{R_E^3}}$$

(b) For small differences,

$$\frac{\left|\Delta g\right|}{\Delta r} = \frac{\left|\Delta g\right|}{h} = \frac{2GM_E}{R_E^3}$$

Thus,

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$$\boxed{|\Delta g| = \frac{2GM_E h}{R_E^3}}$$

(c) $$|\Delta g| = \frac{2\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)(6.00\ \text{m})}{\left(6.37\times10^{6}\ \text{m}\right)^3}$$

$$= \boxed{1.85\times10^{-5}\ \text{m/s}^2}$$

P13.63 (a) Each bit of mass dm in the ring is at the same distance from the object at A. The separate contributions $-\frac{Gmdm}{r}$ to the system energy add up to $-\frac{GmM_{\text{ring}}}{r}$. When the object is at A, this is

$$-\frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(1\,000\ \text{kg})(2.36\times10^{20}\ \text{kg})}{\sqrt{\left(1.00\times10^{8}\ \text{m}\right)^2+\left(2.00\times10^{8}\ \text{m}\right)^2}} = \boxed{-7.04\ \text{N}}$$

(b) When the object is at the center of the ring, the potential energy of the system is

$$-\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(1\,000\ \text{kg}\right)\left(2.36\times10^{20}\ \text{kg}\right)}{1.00\times10^{8}\ \text{m}}$$

$$= \boxed{-1.57\times10^{5}\ \text{J}}$$

(c) Total energy of the object-ring system is conserved:

$$\left(K+U_g\right)_A = \left(K+U_g\right)_B$$

$$0 - 7.04\times10^{4}\ \text{J} = \frac{1}{2}\left(1\,000\ \text{kg}\right)v_B^2 - 1.57\times10^{5}\ \text{J}$$

$$v_B = \left(\frac{2\times8.70\times10^{4}\ \text{J}}{1\,000\ \text{kg}}\right)^{1/2} = \boxed{13.2\ \text{m/s}}$$

***P13.64** The original orbit radius is

$$r = a = 6.37\times10^{6}\ \text{m} + 500\times10^{3}\ \text{m} = 6.87\times10^{6}\ \text{m}$$

The original energy is

$$E_i = -\frac{GMm}{2a}$$

$$= -\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)\left(10^{4}\ \text{kg}\right)}{2\left(6.87\times10^{6}\ \text{m}\right)}$$

$$= -2.90\times10^{11}\ \text{J}$$

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We assume that the perigee distance in the new orbit is 6.87×10^6 m. Then the major axis is $2a = 6.87 \times 10^6 \text{ m} + 2.00 \times 10^7 \text{ m} = 2.69 \times 10^7 \text{ m}$ and the final energy is

$$\begin{aligned} E_f &= -\frac{GMm}{2a} \\ &= -\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(5.98 \times 10^{24} \text{ kg})(10^4 \text{ kg})}{2.69 \times 10^7 \text{ m}} \\ &= -1.48 \times 10^{11} \text{ J} \end{aligned}$$

The energy input required from the engine is

$$E_f - E_i = -1.48 \times 10^{11} \text{ J} - (-2.90 \times 10^{11} \text{ J}) = \boxed{1.42 \times 10^{11} \text{ J}}$$

P13.65 From the walk, $2\pi r = 25\,000$ m. Thus, the radius of the planet is

$$r = \frac{25\,000 \text{ m}}{2\pi} = 3.98 \times 10^3 \text{ m}$$

From the drop:

$$\Delta y = \frac{1}{2} g t^2 = \frac{1}{2} g (29.2 \text{ s})^2 = 1.40 \text{ m}$$

so,

$$g = \frac{2(1.40 \text{ m})}{(29.2 \text{ s})^2} = 3.28 \times 10^{-3} \text{ m/s}^2 = \frac{MG}{r^2}$$

which gives

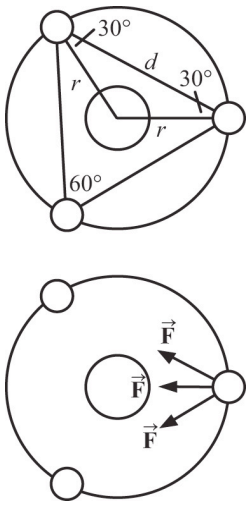
$$M = \boxed{7.79 \times 10^{14} \text{ kg}}$$

P13.66 The distance between the orbiting stars is $d = 2r\cos 30° = \sqrt{3}r$ since $\cos 30° = \frac{\sqrt{3}}{2}$. The net inward force on one orbiting star is

$$\frac{Gmm}{d^2}\cos 30° + \frac{GMm}{r^2} + \frac{Gmm}{d^2}\cos 30° = \frac{mv^2}{r}$$

$$\frac{Gm2\cos 30°}{3r^2} + \frac{GM}{r^2} = \frac{4\pi^2 r^2}{rT^2}$$

$$G\left(\frac{m}{\sqrt{3}} + M\right) = \frac{4\pi^2 r^3}{T^2}$$



ANS. FIG. P13.66

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solving for the period gives

$$T^2 = \frac{4\pi^2 r^3}{G\left(M + m/\sqrt{3}\right)}$$

$$T = 2\pi\left(\frac{r^3}{G\left(M + m/\sqrt{3}\right)}\right)^{1/2}$$

P13.67 (a) We find the period from

$$T = \frac{2\pi r}{v} = \frac{2\pi\left(30\,000 \times 9.46 \times 10^{15}\ \text{m}\right)}{2.50 \times 10^5\ \text{m/s}} = 7 \times 10^{15}\ \text{s}$$

$$= \boxed{2 \times 10^8\ \text{yr}}$$

(b) We estimate the mass of the Milky Way from

$$M = \frac{4\pi^2 a^3}{GT^2} = \frac{4\pi^2\left(30\,000 \times 9.46 \times 10^{15}\ \text{m}\right)^3}{\left(6.67 \times 10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(7.13 \times 10^{15}\ \text{s}\right)^2},$$

$$= 2.66 \times 10^{41}\ \text{kg}$$

or $\boxed{\text{about } 10^{41}\ \text{kg}}$

Note that this is the mass of the galaxy contained within the Sun's orbit of the galactic center. Recent studies show that the true mass of the galaxy, including an extended halo of dark matter, is at least an order of magnitude larger than our estimate.

(c) A solar mass is about 1×10^{30} kg: $10^{41}/10^{30} = 10^{11}$

The number of stars is $\boxed{\text{on the order of } 10^{11}}$.

P13.68 Energy conservation for the two-sphere system from release to contact:

$$-\frac{Gmm}{R} = -\frac{Gmm}{2r} + \frac{1}{2}mv^2 + \frac{1}{2}mv^2$$

$$Gm\left(\frac{1}{2r} - \frac{1}{R}\right) = v^2 \quad \rightarrow \quad v = \left(Gm\left[\frac{1}{2r} - \frac{1}{R}\right]\right)^{1/2}$$

(a) The injected momentum is the final momentum of each sphere,

$$mv = m^{2/2}\left(Gm\left[\frac{1}{2r} - \frac{1}{R}\right]\right)^{1/2} = \boxed{\left[Gm^3\left(\frac{1}{2r} - \frac{1}{R}\right)\right]^{1/2}}$$

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(b) If they now collide elastically each sphere reverses its velocity to receive impulse

$$mv-(-mv)=2mv=\boxed{2\left[Gm^3\left(\frac{1}{2r}-\frac{1}{R}\right)\right]^{1/2}}$$

P13.69 (a) The net torque exerted on the Earth is zero. Therefore, the angular momentum of the Earth is conserved. We use this to find the speed at aphelion:

$$mr_a v_a = mr_p v_p$$

and

$$v_a = v_p\left(\frac{r_p}{r_a}\right)=\left(3.027\times10^4\ \text{m/s}\right)\left(\frac{1.471}{1.521}\right)=\boxed{2.93\times10^4\ \text{m/s}}$$

(b) $$K_p=\frac{1}{2}mv_p^2=\frac{1}{2}\left(5.98\times10^{24}\ \text{kg}\right)\left(3.027\times10^4\ \text{m/s}\right)^2=\boxed{2.74\times10^{33}\ \text{J}}$$

$$U_p=-\frac{GmM}{r_p}$$

$$=-\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)\left(1.99\times10^{30}\ \text{kg}\right)}{1.471\times10^{11}\ \text{m}}$$

$$=\boxed{-5.40\times10^{33}\ \text{J}}$$

(c) Using the same form as in part (b),

$$K_a=\boxed{2.57\times10^{33}\ \text{J}}\ \text{and}\ U_a=\boxed{-5.22\times10^{33}\ \text{J}}$$

(d) Compare to find that

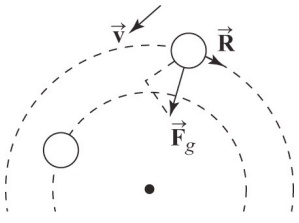
$$K_p+U_p=\boxed{-2.66\times10^{33}\ \text{J}}\ \text{and}\ K_a+U_a=\boxed{-2.65\times10^{33}\ \text{J}}.$$

They agree, with a small rounding error.

P13.70 For both circular orbits,

$$\sum F=ma:\ \frac{GM_E m}{r^2}=\frac{mv^2}{r}$$

$$v=\sqrt{\frac{GM_E}{r}}$$



ANS. FIG. P13.70

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(a) The original speed is

$$v_i = \sqrt{\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)}{6.37\times10^{6}\ \text{m}+2.00\times10^{5}\ \text{m}}}$$

$$= \boxed{7.79\times10^{3}\ \text{m/s}}$$

(b) The final speed is

$$v_i = \sqrt{\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)}{6.47\times10^{6}\ \text{m}}}$$

$$= \boxed{7.85\times10^{3}\ \text{m/s}}$$

The energy of the satellite-Earth system is

$$K+U_g = \frac{1}{2}mv^2 - \frac{GM_Em}{r} = \frac{1}{2}m\frac{GM_E}{r} - \frac{GM_E}{r} = -\frac{GM_Em}{2r}$$

(c) Originally,

$$E_i = -\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)\left(100\ \text{kg}\right)}{2\left(6.57\times10^{6}\ \text{m}\right)}$$

$$= \boxed{-3.04\times10^{9}\ \text{J}}$$

(d) Finally,

$$E_f = -\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg}\right)\left(100\ \text{kg}\right)}{2\left(6.47\times10^{6}\ \text{m}\right)}$$

$$= \boxed{-3.08\times10^{9}\ \text{J}}$$

(e) Thus the object speeds up as it spirals down to the planet. The loss of gravitational energy is so large that the total energy decreases by

$$E_i - E_f = -3.04\times10^{9}\ \text{J} - \left(-3.08\times10^{9}\ \text{J}\right) = \boxed{4.69\times10^{7}\ \text{J}}$$

(f) The only forces on the object are the backward force of air resistance *R*, comparatively very small in magnitude, and the force of gravity. Because the spiral path of the satellite is not perpendicular to the gravitational force,

one component of the gravitational force pulls forward on the satellite

to do positive work and make its speed increase.

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P13.71 The centripetal acceleration of the blob comes from gravitational acceleration:

$$\frac{v^2}{r} = \frac{M_c G}{r^2} = \frac{4\pi^2 r^2}{T^2 r}$$

$$GM_c T^2 = 4\pi^2 r^3$$

Solving for the radius gives

$$r = \left[\frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)(20)\left(1.99\times10^{30}\ \text{kg}\right)\left(5.00\times10^{-3}\ \text{s}\right)^2}{4\pi^2} \right]^{1/3}$$

$$r_{\text{orbit}} = \boxed{119\ \text{km}}$$

P13.72 From Kepler's third law, minimum period means minimum orbit size. The "treetop satellite" in Problem 38 has minimum period. The radius of the satellite's circular orbit is essentially equal to the radius R of the planet.

$$\sum F = ma: \quad \frac{GMm}{R^2} = \frac{mv^2}{R} = \frac{m}{R}\left(\frac{2\pi R}{T}\right)^2$$

$$G\rho V = \frac{R^2\left(4\pi^2 R^2\right)}{RT^2}$$

$$G\rho\left(\frac{4}{3}\pi R^3\right) = \frac{4\pi^2 R^3}{T^2}$$

The radius divides out: $T^2 G\rho = 3\pi \quad \rightarrow \quad \boxed{T = \sqrt{\frac{3\pi}{G\rho}}}$

P13.73 Let m represent the mass of the meteoroid and v_i its speed when far away. No torque acts on the meteoroid, so its angular momentum is conserved as it moves between the distant point and the point where it grazes the Earth, moving perpendicular to the radius:



ANS. FIG. P13.73

$$L_i = L_f: \ m\vec{\mathbf{r}}_i \times \vec{\mathbf{v}}_i = m\vec{\mathbf{r}}_f \times \vec{\mathbf{v}}_f$$

$$m\left(3R_E v_i\right) = mR_E v_f$$

$$v_f = 3v_i$$

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Now, the energy of the meteoroid-Earth system is also conserved:

$$\left(K+U_g\right)_i=\left(K+U_g\right)_f:\ \frac{1}{2}mv_i^2+0=\frac{1}{2}mv_f^2-\frac{GM_Em}{R_E}$$

$$\frac{1}{2}v_i^2=\frac{1}{2}\left(9v_i^2\right)-\frac{GM_E}{R_E}$$

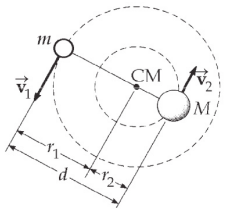
$$\frac{GM_E}{R_E}=4v_i^2: \qquad \boxed{v_i=\sqrt{\frac{GM_E}{4R_E}}}$$

P13.74 If we choose the coordinate of the center of mass at the origin, then

$$0=\frac{\left(Mr_2-mr_1\right)}{M+m} \qquad \text{and} \qquad Mr_2=mr_1$$

(Note: this is equivalent to saying that the net torque must be zero and the two experience no angular acceleration.) For each mass $F = ma$ so

$$mr_1\omega_1^2=\frac{MGm}{d^2} \qquad \text{and} \qquad Mr_2\omega_2^2=\frac{MGm}{d^2}$$



ANS. FIG. P13.74

Combining these two equations and using $d = r_1 + r_2$ gives

$\left(r_1+r_2\right)\omega^2=\dfrac{(M+m)G}{d^2}$ with

$$\omega_1=\omega_2=\omega$$

and

$$T=\frac{2\pi}{\omega}$$

we find

$$\boxed{T^2=\frac{4\pi^2d^3}{G\left(M+m\right)}}$$

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P13.75 The gravitational forces the particles exert on each other are in the x direction. They do not affect the velocity of the center of mass. Energy is conserved for the pair of particles in a reference frame coasting along with their center of mass, and momentum conservation means that the identical particles move toward each other with equal speeds in this frame:

$$U_{gi} + K_i + K_i = U_{gf} + K_f + K_f$$

$$-\frac{Gm_1m_2}{r_i} + 0 = -\frac{Gm_1m_2}{r_f} + \frac{1}{2}m_1v^2 + \frac{1}{2}m_2v^2$$

$$-\frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(1\ 000\ \text{kg})^2}{20.0\ \text{m}}$$

$$= -\frac{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(1\ 000\ \text{kg})^2}{2.00\ \text{m}} + 2\left(\frac{1}{2}\right)(1\ 000\ \text{kg})v^2$$

$$\left(\frac{3.00\times10^{-5}\ \text{J}}{1\ 000\ \text{kg}}\right)^{1/2} = v = 1.73\times10^{-4}\ \text{m/s}$$

Then their vector velocities are $(800 + 1.73\times10^{-4})\,\hat{\mathbf{i}}$ m/s and $(800 - 1.73\times10^{-4})\hat{\mathbf{i}}$ m/s for the trailing particle and the leading particle, respectively.

P13.76 (a) The gravitational force exerted on m by the Earth (mass M_E) accelerates m according to $g_2 = \frac{GM_E}{r^2}$. The equal-magnitude force exerted on the Earth by m produces acceleration of the Earth given by $g_1 = \frac{Gm}{r^2}$. The acceleration of relative approach is then

$$g_2 + g_1 = \frac{Gm}{r^2} + \frac{GM_E}{r^2}$$

$$= \frac{\left(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2\right)\left(5.98\times10^{24}\ \text{kg} + m\right)}{\left(1.20\times10^{7}\ \text{m}\right)^2}$$

$$= \boxed{\left(2.77\ \text{m/s}^2\right)\left(1 + \frac{m}{5.98\times10^{24}\ \text{kg}}\right)}$$

(b) and (c) Here m = 5 kg and m = 2000 kg are both negligible compared to the mass of the Earth, so the acceleration of relative approach is just $\boxed{2.77\ \text{m/s}^2}$.

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(d) Substituting $m = 2.00 \times 10^{24}$ kg into the expression for $(g_1 + g_2)$ above gives

$$g_1 + g_2 = \boxed{3.70 \text{ m/s}^2}$$

(e) Any object with mass small compared to the Earth starts to fall with acceleration 2.77 m/s^2. As m increases to become comparable to the mass of the Earth, the acceleration increases, and can become arbitrarily large. It approaches a direct proportionality to m.

P13.77 For the Earth, $\sum F = ma$: $\dfrac{GM_s m}{r^2} = \dfrac{mv^2}{r} = \dfrac{m}{r}\left(\dfrac{2\pi r}{T}\right)^2$

Then $GM_s T^2 = 4\pi^2 r^3$

Also, the angular momentum $L = mvr = m\dfrac{2\pi r}{T}r$ is a constant for the Earth. We eliminate $r = \sqrt{\dfrac{LT}{2\pi m}}$ between the equations:

$$GM_s T^2 = 4\pi^2\left(\frac{LT}{2\pi m}\right)^{3/2} \quad \text{gives} \quad GM_s T^{1/2} = 4\pi^2\left(\frac{L}{2\pi m}\right)^{3/2}$$

Now the rates of change with time t are described by

$$GM_s\left(\frac{1}{2}T^{-1/2}\frac{dT}{dt}\right) + G\left(1\frac{dM_s}{dt}T^{1/2}\right) = 0$$

or

$$\frac{dT}{dt} = -\frac{dM_s}{dt}\left(2\frac{T}{M_s}\right) \approx \frac{\Delta T}{\Delta t}$$

which gives

$$\Delta T \approx -\Delta t\frac{dM_s}{dt}\left(2\frac{T}{M_s}\right)$$

$$= -(5\,000 \text{ yr})\left(\frac{3.16\times10^7 \text{ s}}{1 \text{ yr}}\right)(-3.64\times10^9 \text{ kg/s}) \times\left(2\frac{1 \text{ yr}}{1.99\times10^{30} \text{ kg}}\right)$$

$$\Delta T = \boxed{5.78\times10^{-10} \text{ s}}$$

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Challenge Problems

P13.78 Let m represent the mass of the spacecraft, r_E the radius of the Earth's orbit, and x the distance from Earth to the spacecraft.

The Sun exerts on the spacecraft a radial inward force of

$$F_s = \frac{GM_s m}{\left(r_E - x\right)^2}$$

while the Earth exerts on it a radial outward force of

$$F_E = \frac{GM_E m}{x^2}$$

The net force on the spacecraft must produce the correct centripetal acceleration for it to have an orbital period of 1.000 year.

$$\text{Thus, } F_S - F_E = \frac{GM_S m}{\left(r_E - x\right)^2} - \frac{GM_E m}{x^2} = \frac{mv^2}{\left(r_E - x\right)} = \frac{m}{\left(r_E - x\right)}\left[\frac{2\pi\left(r_E - x\right)}{T}\right]^2$$

$$\text{which reduces to} \qquad \frac{GM_S}{\left(r_E - x\right)^2} - \frac{GM_E}{x^2} = \frac{4\pi^2\left(r_E - x\right)}{T^2} \qquad \textbf{[1]}$$

Cleared of fractions, this equation would contain powers of x ranging from the fifth to the zeroth. We do not solve it algebraically. We may test the assertion that x is 1.48×10^9 m by substituting it into the equation, along with the following data: $M_S = 1.99 \times 10^{30}$ kg, $M_E = 5.974 \times 10^{24}$ kg, $r_E = 1.496 \times 10^{11}$ m, and $T = 1.000 \text{ yr} = 3.156 \times 10^7$ s.

With $x = 1.48 \times 10^9$ m, the result is

$$6.053 \times 10^{-3} \text{ m/s}^2 - 1.82 \times 10^{-3} \text{ m/s}^2 \approx 5.870\,8 \times 10^{-3} \text{ m/s}^2$$

or

$$5.870\,9 \times 10^{-3} \text{ m/s}^2 \approx 5.870\,8 \times 10^{-3} \text{ m/s}^2$$

To three-digit precision, the solution is 1.48×10^9 m.

As an equation of fifth degree, equation [1] has five roots. The Sun-Earth system has five Lagrange points, all revolving around the Sun synchronously with the Earth. The SOHO and ACE satellites are at one. Another is beyond the far side of the Sun. Another is beyond the night side of the Earth. Two more are on the Earth's orbit, ahead of the planet and behind it by 60°. The twin satellites of NASA's STEREO mission, giving three-dimensional views of the Sun from orbital positions ahead of and trailing Earth, passed through these Lagrange points in 2009. The Greek and Trojan asteroids are at the co-orbital Lagrange points of the Jupiter-Sun system.

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P13.79 (a) From the data about perigee, the energy of the satellite-Earth system is

$$E = \frac{1}{2}mv_p^2 - \frac{GM_E m}{r_p} = \frac{1}{2}(1.60\text{ kg})(8.23\times10^3\text{ m/s})^2 - \frac{(6.67\times10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\text{ kg})(1.60\text{ kg})}{7.02\times10^6\text{ m}}$$

or $E = \boxed{-3.67\times10^7\text{ J}}$

(b) $L = mvr\sin\theta = mv_p r_p \sin 90.0°$

$= (1.60\text{ kg})(8.23\times10^3\text{ m/s})(7.02\times10^6\text{ m})$

$= \boxed{9.24\times10^{10}\text{ kg}\cdot\text{m}^2/\text{s}}$

(c) Since both the energy of the satellite-Earth system and the angular momentum of the Earth are conserved,

at apogee we must have

$$\frac{1}{2}mv_a^2 - \frac{GMm}{r_a} = E$$

and $mv_a r_a \sin 90.0° = L$

Thus,

$$\frac{1}{2}(1.60\text{ kg})v_a^2 - \frac{(6.67\times10^{-11}\text{ N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\text{ kg})(1.60\text{ kg})}{r_a} = -3.67\times10^7\text{ J}$$

and $(1.60\text{ kg})v_a r_a = 9.24\times10^{10}\text{ kg}\cdot\text{m}^2/\text{s}$

Solving simultaneously, and suppressing units,

$$\frac{1}{2}(1.60)v_a^2 - \frac{(6.67\times10^{-11})(5.98\times10^{24})(1.60)(1.60)v_a}{9.24\times10^{10}} = -3.67\times10^7$$

which reduces to

$$0.800v_a^2 - 11\,046v_a + 3.672\,3\times10^7 = 0$$

so $$v_a = \frac{11\,046 \pm \sqrt{(11\,046)^2 - 4(0.800)(3.672\,3\times10^7)}}{2(0.800)}$$

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This gives $v_a = 8\,230$ m/s or $\boxed{5\,580 \text{ m/s}}$. The smaller answer refers to the velocity at the apogee while the larger refers to perigee.

Thus,

$$r_a = \frac{L}{mv_a} = \frac{9.24\times10^{10}\ \text{kg}\cdot\text{m}^2/\text{s}}{(1.60\ \text{kg})(5.58\times10^{3}\ \text{m/s})} = \boxed{1.04\times10^{7}\ \text{m}}$$

(d) The major axis is $2a = r_p + r_a$, so the semimajor axis is

$$a = \frac{1}{2}(7.02\times10^{6}\ \text{m} + 1.04\times10^{7}\ \text{m}) = \boxed{8.69\times10^{6}\ \text{m}}$$

(e) $$T = \sqrt{\frac{4\pi^2 a^3}{GM_E}} = \sqrt{\frac{4\pi^2(8.69\times10^{6}\ \text{m})^3}{(6.67\times10^{-11}\ \text{N}\cdot\text{m}^2/\text{kg}^2)(5.98\times10^{24}\ \text{kg})}}$$

$T = 8\,060$ s = $\boxed{134 \text{ min}}$

***P13.80** (a) Energy of the spacecraft-Mars system is conserved as the spacecraft moves between a very distant point and the point of closest approach:

$$0 + 0 = \frac{1}{2}mv_r^2 - \frac{GM_{\text{Mars}}m}{r}$$

$$v_r = \sqrt{\frac{2GM_{\text{Mars}}}{r}}$$

After the engine burn, for a circular orbit we have

$$\sum F = ma: \qquad \frac{GM_{\text{Mars}}m}{r^2} = \frac{mv_0^2}{r}$$

$$v_0 = \sqrt{\frac{GM_{\text{Mars}}}{r}}$$

The percentage reduction from the original speed is

$$\frac{v_r - v_0}{v_r} = \frac{\sqrt{2}v_0 - v_0}{\sqrt{2}v_0} = \frac{\sqrt{2}-1}{\sqrt{2}}\times100\% = \boxed{29.3\%}$$

(b) The answer to part (a) applies with $\boxed{\text{no changes}}$, as the solution to part (a) shows.

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ANSWERS TO EVEN-NUMBERED PROBLEMS

P13.2 $\sim 10^{-7}$ N

P13.4 (a) 4.39×10^{20} N; (b) 1.99×10^{20} N; (c) 3.55×10^{22} N; (d) The force exerted by the Sun on the Moon is much stronger than the force of the Earth on the Moon.

P13.6 $\left(-10.0\hat{\mathbf{i}} + 5.93\hat{\mathbf{j}}\right) \times 10^{-11}$ N

P13.8 The situation is impossible because no known element could compose the spheres.

P13.10 3.06×10^{-8} m

P13.12 $\frac{2}{3}$

P13.14 (a) $\frac{2MGr}{\left(r^2 + a^2\right)^{3/2}}$ toward the center of mass; (b) At $r = 0$, the fields of the two objects are equal in magnitude and opposite in direction, to add to zero; (c) As $r \to 0, 2MGr\left(r^2 + a^2\right)^{-3/2}$ approaches $2MG(0)/a^3 = 0$; (d) When r is much greater than a, the angles the field vectors make with the x axis become smaller. At very great distances, the field vectors are almost parallel to the axis; therefore they begin to look like the field vector from a single object of mass $2M$; (e) As r becomes much larger than a, the expression approaches $2MGr\left(r^2 + 0^2\right)^{-3/2} = 2MGr/r^3 = 2MG/r^2$ as required.

P13.16 (a) 1.31×10^{17} N; (b) 2.62×10^{12} N/kg

P13.18 1.90×10^{27} kg

P13.20 (a) The particle does posses angular momentum because it is not headed straight for the origin. (b) Its angular momentum is constant. There are no identified outside influences acting on the object. (c) See P13.20(c) for full explanation.

P13.22 1.30 revolutions

P13.24 1.27

P13.26 1.63×10^{4} rad/s

P13.28 (a) 6.02×10^{24} kg; (b) The Earth wobbles a bit as the Moon orbits it, so both objects move nearly in circles about their center of mass, staying on opposite sides of it.

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P13.30 (a) -4.77×10^9 J; (b) 569 N

P13.32 4.17×10^{10} J

P13.34 (a) See P13.34 for full description; (b) 340 s

P13.36 1.66×10^4 m/s

P13.38 $\sqrt{2}v$

P13.40 (a) 0.980; (b) 127 yr; (c) -2.13×10^{17} J

P13.42 $\dfrac{GM_E m}{12R_E}$

P13.44 (a) 42.1 km/s; (b) 2.20×10^{11} m

P13.46 (a) $2\pi\sqrt{\dfrac{(R_E+h)^3}{GM_E}}$; (b) $\sqrt{\dfrac{GM_E}{R_E+h}}$; (c) $GM_E m\left[\dfrac{R_E+2h}{2R_E(R_E+h)}\right]-\dfrac{2\pi^2 R_E^2 m}{(86\ 400\ \text{s})^2}$

P13.48 (a) $v_i=\left(\dfrac{GM_E}{r}\right)^{1/2}$; (b) $\dfrac{5}{4}\left(\dfrac{GM_E}{r}\right)^{1/2}$; (c) $r_f=\dfrac{25r}{7}$

P13.50 (a) 3.07×10^6 m; (b) the rocket would travel farther from Earth

P13.52 492 m/s

P13.54 If one uses the result $v=\sqrt{\dfrac{GM}{r}}$ and the relation $v=(2\pi\tau/T),$ one finds the radius of the orbit to be smaller than the radius of the Earth, so the spacecraft would need to be in orbit underground.

P13.56 (a) 0.700 rad/s; (b) Because his feet stay in place on the floor, his head will be moving at the same tangential speed as his feet. However, his feet and his head are travelling in circles of different radii; (c) If he's not careful, there could be a collision between his head and the wall (see P13.56 for full explanation)

P13.58 (a) $h=\dfrac{R_E v_i^2}{v_{esc}^2-v_i^2}$; (b) $v_f=v_{esc}\sqrt{\dfrac{h}{R_E+h}}$; (c) With $v_1 << v_{esc}, h\approx\dfrac{R_E v_i^2}{v_{esc}^2}=\dfrac{R_E v_i^2 R_E}{2GM_E}$. But $g=\dfrac{GM_E}{R_E^2},$ so $h=\dfrac{v_i^2}{2g}$ in agreement with $0^2=v_i^2+2(-g)(h-0)$.

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P13.60 (a) The two appropriate isolated system models are conservation of momentum and conservation of energy applied to the system consisting of the two spheres; (b) $-2\vec{\mathbf{v}}_{2f}$; (c) $\sqrt{\frac{2GM}{3R} - 2v_{2f}^2}$;

(d) $v_2 = \frac{1}{3}\sqrt{G\frac{M}{R}}$, $v_1 = \frac{2}{3}\sqrt{G\frac{M}{R}}$

P13.62 (a) $\frac{dg}{dr} = -\frac{2GM_E}{R_E^3}$; (b) $|\Delta g| = \frac{2GM_E h}{R_E^3}$; (c) $1.85 \times 10^{-5}\ \text{m/s}^2$

P13.64 1.42×10^{11} J

P13.66 See P13.66 for the full answer.

P13.68 (a) $mv = \left[GM^3\left(\frac{1}{2r} - \frac{1}{R}\right)\right]^{1/2}$; (b) $2\left[GM^3\left(\frac{1}{2r} - \frac{1}{R}\right)\right]^{1/2}$

P13.70 (a) 7.79×10^3 m/s; (b) 7.85×10^3 m/s; (c) -3.04×10^9 J; (d) -3.08×10^9 J; (e) 4.69×10^7 J; (f) one component of the gravitational force pulls forward on the satellite

P13.72 See P13.72 for full description.

P13.74 See P13.74 for full description.

P13.76 (a) $\left(2.77\ \text{m/s}^2\right)\left(1 + \frac{m}{5.98 \times 10^{24}\ \text{kg}}\right)$; (b and c) $2.77\ \text{m/s}^2$; (d) $3.70\ \text{m/s}^2$;

(e) Any object with mass small compared to the Earth starts to fall with acceleration $2.77\ \text{m/s}^2$. As m increases to become comparable to the mass of the Earth, the acceleration increases and can become arbitrarily large. It approaches a direct proportionality to m.

P13.78 See P13.78 for full description.

P13.80 (a) 29.3%; (b) no changes

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